UNSA
UNIVERSIDAD NACIONAL DE SAN AGUSTÍN DE AREQUIPA

DECLARACIÓN JURADA DE COTIZACION (BIENES)

DATOS DE PROVEEDOR	
Razón Social	SERCIA EIRL
RUC	2060469242
Dirección	CALLE HUANTA 111 URB TASAHUAYO
Teléfonos	956337881
Email	VENTAS@SERCIAGROUP.COM SERCIAASESORESYCONSULTORES@GMAIL.COM
Vigencia de la oferta	10 DIAS

DATOS DE CONTACTO	
Nombre	EMERSON VALDIVIA
Cargo	EJECUTIVO COMERCIAL
Teléfono Movil	956337881
Fecha	9/8/2026

DATOS DEL BIEN O BIENES SOLICITADOS				PROPUESTA ECONOMICA							
				OFERTA TÉCNICA						OFERTA ECONÓMICA	
N°	DESCRIPCION	U.M.	Cantidad	Marca	Modelo	Garantía (Meses)	Plazo de entrega (Dias Calendario)	Cumple RTM y Condiciones (SI/NO)	Observaciones (Indicar de forma detallada)	PRECIO UNITARIO INCLUIDO IGV	TOTAL
1	controloador	Unidad	1	Arduino	Nano 33 IoT	12	10	si	(SKU ABX00027)	160	160.00
2	dispositivo de comunicación	Unidad	1	Teltonika	RUT951	12	10	si		1360	1,360.00
3	Sensor de Calidad del Aire	Unidad	1	Sensirion	SPS30	12	10	si		165	165.00
4	Sensor de Luminancia	Unidad	1	nacional	TSL2591	12	10	si	✓ Rango 0.1 a 40,000 Lux con cambio automático ✓ 3 direcciones I²C seleccionables (0x29 / 0x39 / 0x49) ✓ Resolución 16 bits — precisión 0.003 Lux/Count ✓ Doble fotodiodo: visible + infrarrojo ✓ Standby de solo 15µA — ideal para baterías ✓ Permite múltiples sensores en el mismo bus I²C	55	55.00
5	Sensor de Radiacion	Unidad	1	DFRobot	SEN0463	12	30	si	importacion	420	420.00
6	Sensor de Ruido Ambiental	Unidad	1	nacional	INMP441	12	10	si	Interfaz de salida: I2S digital con datos de 24 bits Relación señal-ruido (SNR): 61 dBA Sensibilidad: -26 dBFS Respuesta en frecuencia: de 60 Hz a 15 kHz Consumo de corriente: 1.4 mA Nivel de fuente de alimentación: -75 dBFS Voltaje de operación: 1.8 V a 3.3 V Tipo de micrófono: MEMS omnidireccional Puerto acústico: Inferior Aplicaciones: Grabación de voz, asistentes inteligentes, IoT, sistemas embebidos	51	51.00
7	Sensor de Temperatura y Humedad Ambiental	Unidad	1	nacional	SHT45	12	10	si		179	179.00
8	Sensor Pluviómetro	Unidad	1	Rika Sensors	RK400-04	12	10	si		1295	1,295.00
										TOTAL S/.	3,685.00

Lugar de entrega en	Almacén Central de la U N S A

Nota: No se aceptan cambios ni devoluciones salvo producto defectuoso

(*) Declaro bajo juramento que la información que proporciono tiene carácter de declaración Jurada al amparo del principio de Veracidad señalado en el Art. IV, numeral 1.7 del Título preliminar y lo dispuesto en el Art. 42 de la Ley de procedimiento Administrativo General – Ley N° 27444, Así mismo, declaro tener conocimiento que si lo Declarado es falso, estoy sujeto a los alcances de las disposiciones Administrativas, Penales y Civiles que correspondan. .

SERCIA CONSULTORES SAC
EMERSON VALDIVIA CARTAGENA
GERENTE GENERAL

FIRMA Y SELLO DEL PROVEEDOR

Nota: No se aceptan cambios ni devoluciones salvo producto defectuoso

TELTONIKA | Networks

RUT951

Quick Start Guide v1.0

Wiki knowledge base

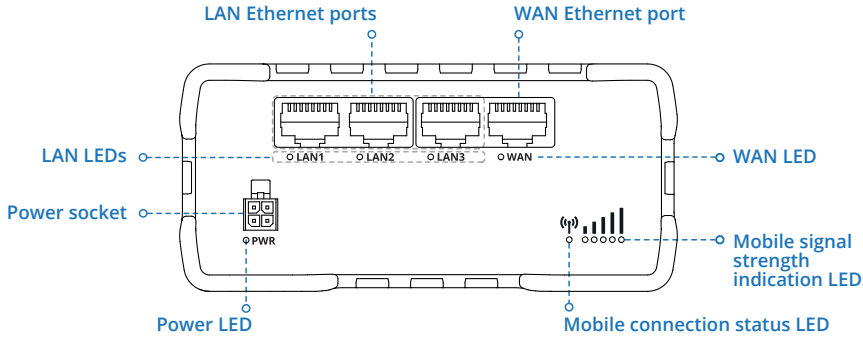
https://wiki.teltonika-networks.com/



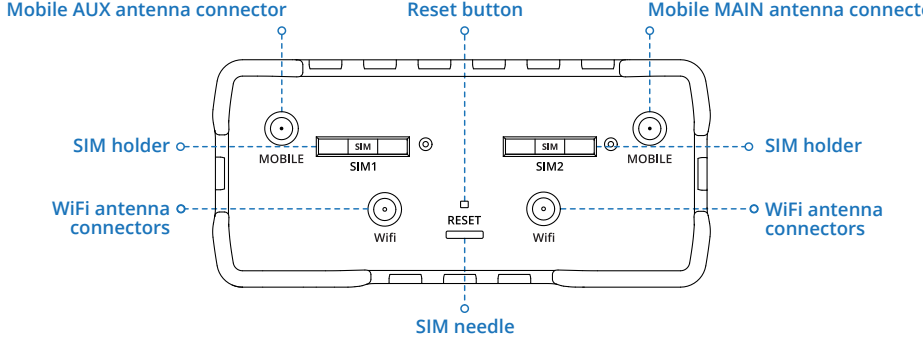




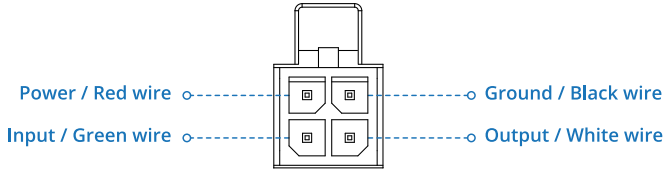
FRONT VIEW



BACK VIEW

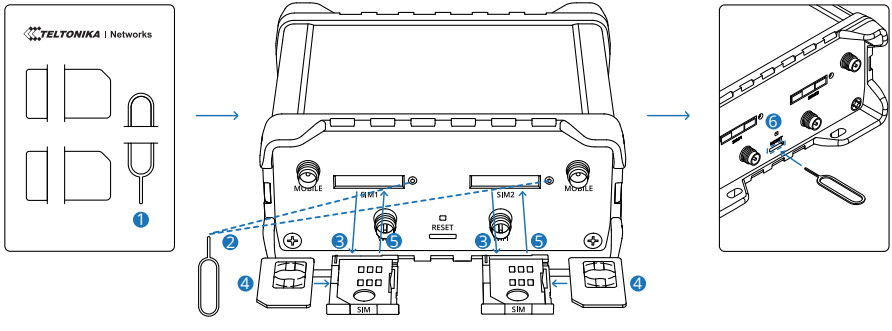


POWER SOCKET PINOUT



HARDWARE INSTALLATION

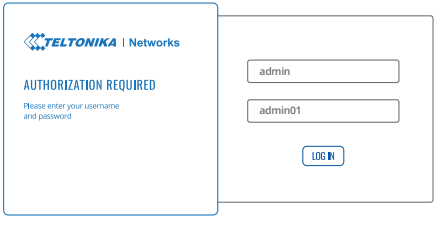
1. Pull out the SIM needle from SIM holder kit.
2. Push the SIM holder kit button with the SIM needle.
3. Pull out the SIM holder.
4. Insert your SIM card into the SIM holder.
5. Slide the SIM holder back into the router.
6. You may store the SIM needle in the designated holder for future use.
7. Attach all antennas.
8. Connect the power adapter to the socket on the front of the device. Then plug the other end of the power adapter into a power outlet.
9. Connect to the device wirelessly using SSID and password provided on the device information label or use an Ethernet cable connected to LAN port.



LOGIN TO DEVICE

1. To enter the router's Web interface (WebUI), type **http://192.168.1.1** into the URL field of your Internet browser*.
2. Use login information shown in image A when prompted for authentication.
3. After you login, you will be prompted to change your password for security reasons. The new password must contain at least 8 characters, including at least one uppercase letter, one lowercase letter and one digit. This step is mandatory and **you will not be able to interact with the router's WebUI before you change the password**.
4. When you change the router's password, the **Configuration Wizard** will start. The **Configuration Wizard** is a tool used to setup some of the router's main operating parameters.
5. Go to the **Overview** page and pay attention to the **Signal Strength** indication (image B). To maximize the cellular performance try adjusting the antennas or changing the location of your device to achieve the best signal conditions.

A.



B.



TECHNICAL INFORMATION

Radio specifications	
RF technologies	2G, 3G, 4G, WiFi
Max RF power	33 dBm@GSM, 24 dBm@WCDMA, 23 dBm@LTE, 20 dBm@ WiFi
Bundled accessories specifications**	
Power adapter	Input: 0.45 A@100-240 VAC, Output: 9 VDC, 1 A, 4-pin plug
Mobile antenna	698~960/1710~2690 MHz, 50 Ω, VSWR<3, gain*** 4 dBi, omnidirectional, SMA male connector
WiFi antenna	2400~2483.5 MHz, 50 Ω, VSWR<2, gain*** 5 dBi, omnidirectional, RP-SMA male connector

*To ensure maximum compatibility please use following latest internet browsing applications: Chrome, Firefox, Edge, Opera, Safari, Android Chrome, Android Firefox.
**Order code dependent.
***Higher gain antenna can be connected to compensate for cable attenuation when a cable is used. The user is responsible for the compliance with the legal regulations.

SAFETY INFORMATION

RUT951 router must be used in compliance with any and all applicable national and international laws and with any special restrictions regulating the utilization of the communication module in prescribed applications and environments.

[BG] Bulgarian	С настоящето, TELTONIKA NETWORKS декларира, че RUT951 е в съответствие със съществените изисквания и другитеприложими разпоредби на Директива 2014/53/ЕС.
[HR] Croatian	Ovim TELTONIKA NETWORKS izjavljuje da je ovaj RUT951 u skladu s bitnim zahtjevima i ostalim relevantnim odredbama Direktive 2014/53/EU.
[CZ] Czech	TELTONIKA NETWORKS tímto prohlašuje, že RUT951 splňuje základní požadavky a všechna příslušná ustanoveni Směrnice 2014/53/EU.
[DK] Danish	Undertegnede TELTONIKA NETWORKS erklærer herved, at følgende udstyr RUT951 overholder de væsentlige krav og øvrige relevante krav i direktiv 2014/53/EU.
[NL] Dutch	Hierbij verklaart TELTONIKA NETWORKS dat het toestel I RUT951 in overeenstemming is met de essentiële eisen en de andere relevante bepalingen van richtlijn 2014/53/EU.
[EN] English	Hereby, TELTONIKA NETWORKS declares that this RUT951 is in compliance with the essential requirements and other relevant provisions of Directive 2014/53/EU.
[EE] Estonian	Käesolevaga kinnitab TELTONIKA NETWORKS seadme RUT951 vastavust direktiivi CE/RED põhinõuetele ja nimetatud direktiivist tulenevatele teistele asjakohastele sätetele.
[FI] Finish	TELTONIKA NETWORKS vakuuttaa täten että RUT951 tyyppinen laite on direktiivin 2014/53/EU oleellisten vaatimusten ja sitä koskevien direktiivin muiden ehtojen mukainen.
[FR] French	TELTONIKA NETWORKS déclare que cette RUT951 est conforme aux exigences essentielles et aux autres dispositions pertinentes de la directive 2014/53/EU.
[DE] German	Hiermit erklärt TELTONIKA NETWORKS, dass dieser RUT951 den grundlegenden Anforderungen und anderen relevanten Bestimmungen der Richtlinie 2014/53/EU entspricht.
[GR] Greek	ΜΕ ΤΗΝ ΠΑΡΟΥΣΑ Ο ΚΑΤΑΣΚΕΥΑΣΤΗΣ TELTONIKA NETWORKS ΔΗΛΩΝΕΙ ΟΤΙ RUT951 ΣΥΜΜΟΡΦΩΝΕΤΑΙ ΠΡΟΣ ΤΙΣ ΟΥΣΙΩΔΕΙΣ ΑΠΑΙΤΗΣΕΙΣ ΚΑΙ ΤΙΣ ΛΟΙΠΕΣ ΣΧΕΤΙΚΕΣ ΔΙΑΤΑΞΕΙΣ ΤΗΣ ΟΔΗΓΙΑΣ 2014/53/EU.
[HU] Hungarian	A TELTONIKA NETWORKS ezzennel kijelenti, hogy a RUT951 típusú berendezés teljesíti az alapvető követelményeket és más CE/RED irányelvben meghatározott vonatkozó rendelkezéseket.
[IE] Irish	Leis seo, dearbhaíonn TELTONIKA NETWORKS go gcomhlíonann an RUT951 seo bunriachtanais agus forálacha ábhartha eile Threoir 2014/53/EU.
[IT] Italian	Con la presente, TELTONIKA NETWORKS dichiara che questo RUT951 è conforme ai requisiti essenziali e ad altre disposizioni pertinenti della Direttiva 2014/53/EU.
[LV] Latvian	Ar šo TELTONIKA NETWORKS deklarē, ka RUT951 atbilst Direktīvas 2014/53/EU būtiskajām prasībām un citiem ar to saistītajiem noteikumiem.
[LT] Lithuanian	Šiuo dokumentu UAB TELTONIKA NETWORKS deklaruoja, kad šis RUT951 atitinka esminius reikalavimus ir kitas 2014/53/EU Direktyvos nuostatas.
[MT] Maltese	Hawnhekk, TELTONIKA NETWORKS tiddikjara li dan l-RUT951 huwa konformi mar-rekwiżiti essenzjali u dispożizzjonijiet rilevanti oħra tad-Direttiva 2014/53/EU.
[NO] Norwegian	TELTONIKA NETWORKS Erklærer herved at RUT951 er i samsvar med de grunnleggende krav og øvrige relevante krav i direktiv 2014/53/EU.
[PL] Polish	Niniejszym TELTONIKA NETWORKS deklaruje że RUT951 jest zgodny z zasadniczymi wymaganiami i innymi właściwymi postanowieniami Dyrektywy 2014/53/EU.
[PT] Portuguese	Eu, TELTONIKA NETWORKS declaro que o RUT951 cumpre os requisitos essenciais e outras provisões relevantes da Directiva 2014/53/EU.
[RO] Romanian	Prin prezenta, TELTONIKA NETWORKS declară că aparatul RUT951 este în conformitate cu cerințele esențiale și cu alte prevederi pertinente ale Directivei 2014/53/EU.
[SK] Slovak	Spoločnosť TELTONIKA NETWORKS týmto vyhlasuje, že smerovač RUT951 je v súlade so základnými požiadavkami a ďalšími príslušnými ustanoveniami smernice 2014/53/EU.
[SI] Slovenian	TELTONIKA NETWORKS izjavlja, da je ta RUT951 v skladu z bistvenimi zahtevami in drugimi relevantnimi določili direktive 2014/53/EU.
[ES] Spanish	Por la presente, TELTONIKA NETWORKS declara que este RUT951 cumple con los requisitos esenciales y otras exigencias relevantes de la Directiva 2014/53/EU.
[SE] Swedish	Härmed intygar TELTONIKA NETWORKS att denna RUT951 är i överensstämmelse med de väsentliga egenskapskrav och övriga relevanta bestämmelser som framgår av direktiv 2014/53/EU.

The full text of the EU declaration of conformity is available at the following Internet address: https://wiki.teltonika-networks.com/view/RUT951_CE/RED

[BG] Инструкция: Свържете адаптера на захранване, за да включите устройството. Отворете 192.168.1.1 в уеб браузър, за да настроите продукта. Повече информация в https://wiki.teltonika-networks.com/
[HR] Upute za uporabu: Priključite napajanje. Otvorite 192.168.1.1 u svom internet pregledniku kako biste konfigurirali uređaj. Više informacija na https://wiki.teltonika-networks.com
[CZ] Návod k použití: Připojte napájecí adaptér k zapnutí přístroje. Otevřete 192.168.1.1 ve webovém prohlížeči pro konfiguraci produktu. Více informací najdete v https://wiki.teltonika-networks.com/
[DK] Instruktionsbog: Tilslut strømadapteren for at tænde for enheden. Åbn 192.168.1.1 i en webbrowser til at konfigurere produktet. Mere information i https://wiki.teltonika-networks.com/
[NL] Handleiding: Sluit voedingsadapter aan op het apparaat in te schakelen. Open 192.168.1.1 in een webbrowser om het product te configureren. Meer informatie in https://wiki.teltonika-networks.com/
[EN] Instruction Manual: Connect the power adapter to turn on the device. Open 192.168.1.1 in your web browser to configure it. More information on https://wiki.teltonika-networks.com/
[EE] Kasutusjuhend: Ühendage toiteadapter seadme sisselülitamiseks. Avatud 192.168.1.1 in veebilehitseja seadistada toodet. Rohkem teavet https://wiki.teltonika-networks.com/
[FI] Käyttöohje: Kytke virtalähde päälle laitteen. Avaa 192.168.1.1 in selaimen määrittää tuotteen. Lisää tietoa https://wiki.teltonika-networks.com/
[FR] Mode d'emploi: Connectez l'adaptateur d'alimentation pour allumer l'appareil. Ouvrez 192.168.1.1 dans un navigateur Web pour configurer le produit. Plus d'informations dans https://wiki.teltonika-networks.com/
[DE] Bedienungsanleitung: Verbinden Sie das Netzteil, um das Gerät einzuschalten. Öffnen Sie 192.168.1.1 in einem Web Browser, um das Produkt zu konfigurieren. Weitere Informationen im https://wiki.teltonika-networks.com/
[GR] εγχειρίδιο οδηγιών: Συνδέστε τον προσαρμογέα τροφοδοσίας για να ενεργοποιήσετε τη μονάδα. Ανοίξτε 192.168.1.1 σε ένα πρόγραμμα περιήγησης στο Web για να διαμορφώσετε το προϊόν. Περισσότερες πληροφορίες στο https://wiki.teltonika-networks.com/
[HU] Használati utasítás: Csatlakoztassa a hálózati adaptert a készülék bekapcsolásához. Megnyitása 192.168.1.1 egy webböngészőben beállítani a terméket. Több információ https://wiki.teltonika-networks.com/
[IE] Lámhleabhar Treoracha: Ceangail an t-adapter cumhachta chun an gléas a chasadh air. Oscail 192.168.1.1 i do bhrabhsálaí gréasáin chun é a chumrú. Tuilleadh eolais ar https://wiki.teltonika-networks.com
[IT] Manuale di istruzione: collegare l'adattatore di alimentazione per accendere il dispositivo. Apri 192.168.1.1 nel tuo browser web per configurarlo. Maggiori informazioni su https://wiki.teltonika-networks.com
[LV] Lietošanas instrukcija: Pievienojiet Strāvas adapteri, lai ieslēgtu ierīci. Atvērt 192.168.1.1 ar interneta pārlūku, lai konfigurētu produktu. Plašāka informācija https://wiki.teltonika-networks.com/
[LT] Naudojimosi instrukcija: Prijunkite maitinimo adapterį, įjunkite įrenginį. Interneto naršyklėje atidarykite adresą 192.168.1.1 - galite konfigūruoti gaminį. Daugiau informacijos rasite https://wiki.teltonika-networks.com/
[MT] Manwal tal-Istruzzjoni: Qabbad l-adapter tal-enerġija biex tixgħel l-apparat. Iftaħ 192.168.1.1 fil-web browser tiegħek biex tikkonfigurah. Aktar informazzjoni dwar https://wiki.teltonika-networks.com
[NO] Bruksanvisningen: Koble strømadapteren for å slå på enheten. Åpne 192.168.1.1 i en nettleser for å konfigurere produktet. Mer informasjon på https://wiki.teltonika-networks.com/
[PL] Instrukcja obsługi: Podłącz adapter zasilania, aby włączyć urządzenie. Otwórz 192.168.1.1 w przeglądarce internetowej, aby skonfigurować urządzenie. Więcej informacji w https://wiki.teltonika-networks.com/
[PT] Manual de instruções: Conecte o adaptador de alimentação para ligar o aparelho. Abra 192.168.1.1 em um navegador da web para configurar o produto. Mais informações em https://wiki.teltonika-networks.com/
[RO] Instrucțiuni de utilizare: Conectați adaptorul de alimentare pentru a porni aparatul. Deschide 192.168.1.1 întrun browser web pentru a configura produsul. Mai multe informații în https://wiki.teltonika-networks.com/
[SK] Návod na použitie: Pripojením napájacieho adaptéra zapnite zariadenie. Otvorte vo svojom webovom prehliadači stránku s IP adresou 192.168.1.1 a nakonfigurujte ho. Viac informácií na https://wiki.teltonika-networks.com
[SI] Navodila: Priključite napajalnik za vklop naprave. Odprite 192.168.1.1 v spletnem brskalniku nastaviti izdelek. Več informacij v https://wiki.teltonika-networks.com
[ES] Manual de instrucciones: Conecte el adaptador de alimentación para encender la unidad. Abra 192.168.1.1 en un navegador web para configurar el producto. Más información en https://wiki.teltonika-networks.com/
[SE] Instruktionsmanual: Anslut nätadap tern för att slå på enheten. Öppna 192.168.1.1 i en webbläsare för att konfigurera produkten. Mer information på https://wiki.teltonika-networks.com/

This sign means that all used electronic and electric equipment should not be mixed with general household waste.

This sign means that product meets the requirements of the applicable EU directives.

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SENSIRION
THE SENSOR COMPANY

Datasheet SPS30

Particulate Matter Sensor for Air Quality Monitoring and Control

- Unique long-term stability
- Advanced particle size binning
- Superior precision in mass concentration and number concentration sensing
- Small, ultra-slim package
- Fully calibrated digital output



Product Summary
The SPS30 Particulate Matter (PM) sensor is a technological breakthrough in optical PM sensors. Its measurement principle is based on laser scattering and makes use of Sensirion's innovative contamination-resistance technology. This technology, together with high-quality and long-lasting components, enables precise measurements from its first operation and throughout its lifetime of more than ten years. In addition, Sensirion's advanced algorithms provide superior precision for different PM types and higher-resolution particle size binning, opening up new possibilities for the detection of different sorts of environmental dust and other particles. With dimensions of only 41 x 41 x 12 mm^3, it is also the perfect solution for applications where size is of paramount importance, such as wall-mounted or compact air quality devices.

Content

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1 Particulate Matter Sensor Specifications

1.1 Specification Overview

Parameter	Conditions		Value	Units
Mass concentration range	-		0 to 1'000	µg/m³
Mass concentration size range	PM1.0		0.3 to 1.0	µm
	PM2.5		0.3 to 2.5	µm
	PM4		0.3 to 4.0	µm
	PM10		0.3 to 10.0	µm
Mass concentration precision[1,2] for PM1 and PM2.5[3]	0 to 100 µg/m³		± [5 µg/m³ + 5 % m.v.]	
	100 to 1000 µg/m³		±10	% m.v.
Mass concentration precision[1,2] for PM4, PM10[4]	0 to 100 µg/m³		±25	µg/m³
	100 to 1000 µg/m³		±25	% m.v.
Maximum long-term mass concentration precision limit drift	0 to 100 µg/m³		±1.25	µg/m³ / year
	100 to 1000 µg/m³		±1.25	% m.v. / year
Number concentration range	-		0 to 3'000	#/cm³
Number concentration size range	PM0.5		0.3 to 0.5	µm
	PM1.0		0.3 to 1.0	µm
	PM2.5		0.3 to 2.5	µm
	PM4		0.3 to 4.0	µm
	PM10		0.3 to 10.0	µm
Number concentration precision[1,2] for PM0.5, PM1 and PM2.5[3]	0 to 1000 #/cm³		±100	#/cm³
	1000 to 3000 #/cm³		±10	% m.v.
Number concentration precision[1,2] for PM4, PM10[4]	0 to 1000 #/cm³		±250	#/cm³
	1000 to 3000 #/cm³		±25	% m.v.
Maximum long-term number concentration precision limit drift[2]	0 to 1000 #/cm³		±12.5	#/cm³ / year
	1000 to 3000 #/cm³		±1.25	% m.v. / year
Sampling interval	-		1±0.04	s
Typical start-up time[5]	number concentration	200 – 3000 #/cm³	8	s
		100 – 200 #/cm³	16	s
		50 – 100 #/cm³	30	s
Sensor output characteristics	PM2.5 mass concentration		Calibrated to TSI DustTrak™ DRX 8533 Ambient Mode	
	PM2.5 number concentration		Calibrated to TSI OPS 3330	
Lifetime[6]	24 h/day operation		> 10	years
Acoustic emission level	0.2 m	max.	25	dB(A)
Long term acoustic emission level drift	0.2 m	max.	+0.5	dB(A) / year
Additional T-dependent mass and number concentration precision limit drift[2]	temperature difference to 25°C	typ.	±0.5	% m.v. / °C
Weight	-		26.3 ±0.3	g

[1] Also referred to as "between-parts variation" or "device-to-device variation".

[2] For further details, please refer to the document "Sensirion Particulate Matter Sensor Specification Statement".

[3] Verification Aerosol for PM2.5 is a 3% atomized KCl solution. Deviation to reference instrument is verified in end-tests for every sensor after calibration.

[4] PM4 and PM10 output values are calculated based on distribution profile of all measured particles.

[5] Time after starting Measurement-Mode, until a stable measurement is obtained.

[6] Lifetime is based on mean-time-to-failure (MTTF) calculation. Lifetime might vary depending on different operating conditions.

Laser wavelength (DIN EN 60825-1 Class 1)		typ.	660	nm

Table 1: Particulate matter sensor specifications. Default conditions of 25±2 °C, 50±10% relative humidity and 5 V supply voltage apply unless otherwise stated. 'max.' means 'maximum', 'typ.' means 'typical', '% m.v.' means '% of measured value'.

1.2 Recommended Operating Conditions

The sensor shows best performance when operated within recommended normal temperature and humidity range of 10 to 40 °C and 20 to 80 % RH, respectively.

2 Electrical Specifications

2.1 Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Supply voltage	-	4.5	5.0	5.5	V
Supply current	Sleep-Mode	-	38	50	µA
	Idle-Mode	300	330	360	
	Measurement-Mode	45	55	65	mA
	Measurement-Mode, first 200ms (fan start)	-	-	80	
Input high level voltage (V_{IH})	-	2.31	-	5.5	V
Input low level voltage (V_{IL})	-	0	-	0.99	
Output high level voltage (V_{OH})	-	2.9	3.3	3.37	
Output low level voltage (V_{OL})	-	0	0	0.4	

Table 2: Electrical specifications at 25°C.

2.2 Absolute Minimum and Maximum Ratings

Stress levels beyond those listed in Table 3 may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions cannot be guaranteed. Exposure to the absolute maximum rating conditions for extended periods may affect the reliability of the device.

Parameter	Min	Max	Unit
Supply voltage VDD	-0.3	5.5	V
Interface Select SEL	-0.3	4.0	
I/O pins (RX/SDA, TX/SCL)	-0.3	5.5	
Max. current on any I/O pin	-16	16	mA
Operating temperature range	-10	60	°C
Storage temperature range	-40	70	
Operating humidity range	0	95	% RH

Table 3: Absolute minimum and maximum ratings.

2.3 ESD / EMC Ratings

Immunity (Industrial level)

Description	Standard	Rating
Electro Static Discharge	IEC 61000-4-2	±4 kV contact, ±8 kV air
Power-Frequency Magnetic Field	IEC 61000-4-8	30A/m, 50Hz and 60Hz
Radio-Frequency EM-Field AM-modulated	IEC 61000-4-3	80MHz - 1000MHz, 10V/m, 80% AM @1kHz
Radio-Frequency EM-Field AM-modulated	IEC 61000-4-3	1.4GHz – 6GHz, 3V/m, 80% AM @1kHz

Emission (Residential level)

Description	Standard	Rating
Emission in SAC for 30MHz to 230MHz	IEC/CISPR 16	40dB(µV/m) QP @3m
Emission in SAC for 230MHz to 1000MHz	IEC/CISPR 16	47dB(µV/m) QP @3m
Emission in SAC for 1GHz to 3GHz	IEC/CISPR 16	70dB(µV/m) P, 50dB(µV/m) AP @3m
Emission in SAC for 3GHz to 6GHz	IEC/CISPR 16	74dB(µV/m) P, 54dB(µV/m) AP @3m

3 Hardware Interface Specifications

The interface connector is located at the side of the sensor opposite to the air inlet/outlet. Corresponding female plug is ZHR-5 from JST Sales America Inc. In Figure 1 a description of the pin layout is given.

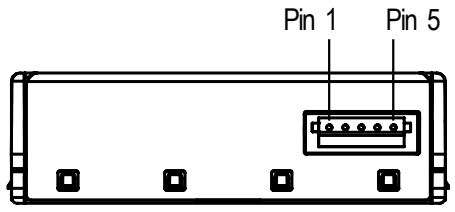


Figure 1: The communication interface connector is located at the side of the sensor opposite to the air outlet.

Pin	Name	Description	Comments
1	VDD	Supply voltage	5V ± 10%
2	RX	UART: Receiving pin for communication	TTL 5V and LVTTL 3.3V compatible
	SDA	I²C: Serial data input / output	
3	TX	UART: Transmitting pin for communication	TTL 5V and LVTTL 3.3V compatible
	SCL	I²C: Serial clock input	
4	SEL	Interface select	Leave floating to select UART
			Pull to GND to select I²C
5	GND	Ground	Housing on GND

Table 4 SPS30 pin assignment.

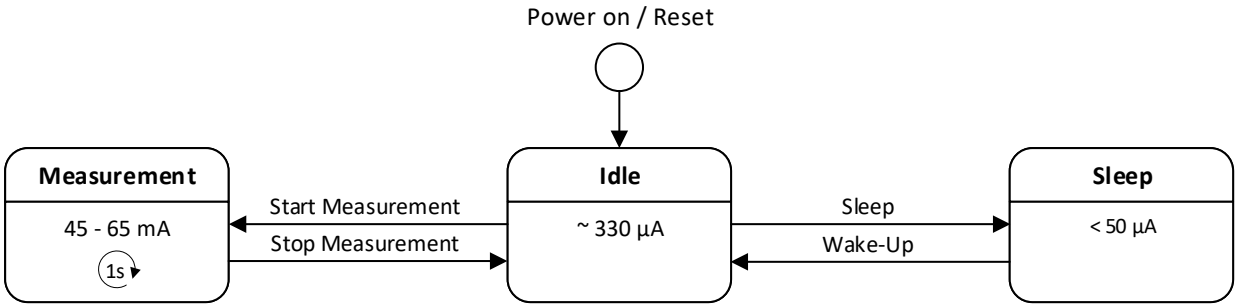
The SPS30 offers both a UART[7] and an I²C interface. For connection cables longer than 20 cm we recommend using the UART interface, due to its intrinsic robustness against electromagnetic interference.

Note, that there is an internal electrical connection between GND pin (5) and metal shielding. Keep this metal shielding electrically floating in order to avoid any unintended currents through this internal connection. If this is not an option, proper external potential equalization between GND pin and any potential connected to the shielding is mandatory. Any current though the connection between GND and metal shielding may damage the product and poses a safety risk through overheating.

[7] Universal Asynchronous Receiver Transmitter.

4 Functional Overview

4.1 Operating Modes



Idle

- After power on or reset the module is in Idle-Mode.
- Most of the internal electronics switched off /reduced power consumption.
- Fan and laser are switched off.
- The module is ready to receive and process any command.

Measurement

- The Measurement-Mode can only be entered from Idle-Mode.
- All electronics switched on / max. power consumption.
- The measurement is running and the module is continuously processing measurement data.
- New readings are available every second.

Sleep

- The Sleep-Mode can only be entered from Idle-Mode.
- Most of the internal electronics switched off / reduced power consumption.
- Fan and laser are switched off.
- Microcontroller is in Sleep-Mode.
- To minimize power consumption, the UART / I2C interface is also disabled.
- A wake-up sequence is needed to turn the module back on. See Wake-up command in the interface description.

4.2 Fan Auto Cleaning

When the module is in Measurement-Mode an automatic fan-cleaning procedure will be triggered periodically following a defined cleaning interval. This will accelerate the fan to maximum speed for 10 seconds in order to blow out the dust accumulated inside the fan.

- Measurement values are not updated while the fan-cleaning is running.
- The default cleaning interval is set to 604’800 seconds (i.e., 168 hours or 1 week) with a tolerance of ±3%.
- The interval can be configured using the Set Automatic Cleaning Interval command.
- Set the interval to 0 to disable the automatic cleaning.
- Once set, the interval is stored permanently in the non-volatile memory.
- If the sensor is switched off, the time counter is reset to 0. Make sure to trigger a cleaning cycle at least every week if the sensor is switched off and on periodically (e.g., once per day).
- The cleaning procedure can also be started manually with the Start Cleaning command.

4.3 Measurement Output Formats

The measurement results can be read with the "Read Measured Values" command. The returned data structure depends on the selected output format. The output format must be specified when stating the measurement with the "Start Measurement command".

IEEE754 float values

Byte #		Datatype	Description
SHDLC	I2C		
0..3	0..5	big-endian float IEEE754	Mass Concentration PM1.0 [µg/m³]
4..7	6..11		Mass Concentration PM2.5 [µg/m³]
8..11	12..17		Mass Concentration PM4.0 [µg/m³]
12..15	18..23		Mass Concentration PM10 [µg/m³]
16..19	24..29		Number Concentration PM0.5 [#/cm³]
20..23	30..35		Number Concentration PM1.0 [#/cm³]
24..27	36..41		Number Concentration PM2.5 [#/cm³]
28..31	42..47		Number Concentration PM4.0 [#/cm³]
32..35	48..53		Number Concentration PM10 [#/cm³]
36..39	54..59		Typical Particle Size[8] [µm]

Unsigned 16-bit integer values[9]

Byte #		Datatype	Description
SHDLC	I2C		
0..1	0..2	big-endian unsigned 16-bit integer	Mass Concentration PM1.0 [µg/m³]
2..3	3..5		Mass Concentration PM2.5 [µg/m³]
4..5	6..8		Mass Concentration PM4.0 [µg/m³]
6..7	9..11		Mass Concentration PM10 [µg/m³]
8..9	12..14		Number Concentration PM0.5 [#/cm³]
10..11	15..17		Number Concentration PM1.0 [#/cm³]
12..13	18..20		Number Concentration PM2.5 [#/cm³]
14..15	21..23		Number Concentration PM4.0 [#/cm³]
16..17	24..26		Number Concentration PM10 [#/cm³]
18..19	27..29		Typical Particle Size[8] [nm]

[8] The typical particle size (TPS) gives an indication on the average particle diameter in the sample aerosol. Such output correlates with the weighted average of the number concentration bins measured with a TSI 3330 optical particle sizer. Consequently, lighter aerosols will have smaller TPS values than heavier aerosols.

[9] Requires at least firmware version 2.0

4.4 Device Status Register

The Device Status Register is a 32-bit register that contains information about the internal state of the module.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										Warning					
res.	res.	res.	res.	res.	res.	res.	res.	res.	res.	SPEED	res.	res.	res.	res.	res.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										Error	Error				
res.	res.	res.	res.	res.	res.	res.	res.	res.	res.	LASER	FAN	res.	res.	res.	res.

Note: All “res.” bits are reserved for internal use or future versions. These bits can be both 0 and 1 and should therefore be ignored.

Bit 21 **SPEED:** Fan speed out of range

0: Fan speed is ok.
1: Fan speed is too high or too low.

- During the first 3 seconds after starting the measurement (fan start-up) the fan speed is not checked.
- The fan speed is also not checked during the auto cleaning procedure.
- Apart from the two exceptions mentioned above, the fan speed is checked once per second in the measurement mode. If it is out of range twice in succession, the SPEED-bit is set.
- At very high or low ambient temperatures, the fan may take longer to reach its target speed after start-up. In this case, the bit will be set. As soon as the target speed is reached, this bit is cleared automatically.
- If this bit is constantly set, this indicates a problem with the power supply or that the fan is no longer working properly.

Bit 5 **LASER:** Laser failure

0: Laser current is ok.
1: Laser is switched on and current is out of range.

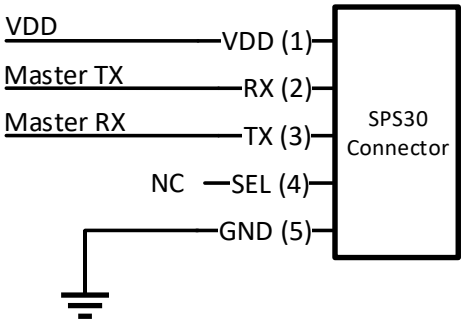
- The laser current is checked once per second in the measurement mode. If it is out of range twice in succession, the LASER-bit is set.
- If the laser current is back within limits, this bit will be cleared automatically.
- A laser failure can occur at very high temperatures outside of specifications or when the laser module is defective.

Bit 4 **FAN:** Fan failure, fan is mechanically blocked or broken.

0: Fan works as expected.
1: Fan is switched on, but the measured fan speed is 0 RPM.

- The fan is checked once per second in the measurement mode. If 0 RPM is measured twice in succession, the FAN bit is set.
- The FAN-bit will not be cleared automatically.
- A fan failure can occur if the fan is mechanically blocked or broken.

5 Operation and Communication through the UART Interface



The following UART settings have to be used:

- Baud Rate: 115'200 bit/s
- Data Bits: 8
- Parity: None
- Stop Bit: 1

Figure 2: Typical UART application circuit.

5.1 Physical Layer

The SPS30 has separate RX and TX lines with unipolar logic levels. See **Figure 3**.

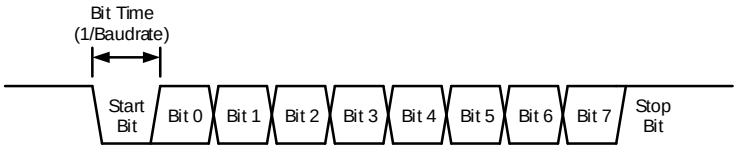


Figure 3: Transmitted byte.

5.2 SHDLC Frame Layer

On top of the UART interface, the SPS30 uses the very powerful and easy-to-implement SHDLC[10] protocol. It is a serial communication protocol based on a master/slave architecture. The SPS30 acts as the slave device.

Data is transferred in logical units called frames. Every transfer is initiated by the master sending a MOSI[11] frame. The slave will respond to the MOSI frame with a slave response, or MISO[12] frame. The two types of frames are shown in **Figure 4**.

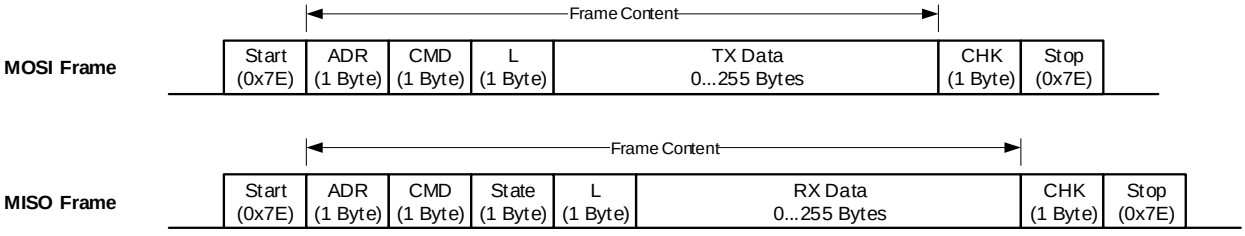


Figure 4: MOSI and MISO frames structure.

[10] Sensirion High-Level Data Link Control.

[11] Master Out Slave In. Frame direction from master to slave.

[12] Master In Slave Out. Frame direction from slave to master.

Start/Stop Byte and Byte-Stuffing

The 0x7E character is sent at the beginning and at the end of the frame to signalize frame start and stop. If this byte (0x7E) occurs anywhere else in the frame, it must be replaced by two other bytes (byte-stuffing). This also applies to the characters 0x7D, 0x11 and 0x13. Use **Table 5** for byte-stuffing.

Original data byte	Transferred data bytes
0x7E	0x7D, 0x5E
0x7D	0x7D, 0x5D
0x11	0x7D, 0x31
0x13	0x7D, 0x33

Table 5 Reference table for byte-stuffing.

Example: Data to send = [0x43, 0x11, 0x7F] → Data transmitted = [0x43, 0x7D, 0x31, 0x7F].

Address

The slave device address is always 0.

Command

In the MOSI frame the command tells the device what to do with the transmitted data. In the MISO frame, the slave just returns the received command.

Length

Length of the “TX Data” or “RX Data” field (before byte-stuffing).

State

The MISO frame contains a state byte, which allows the master to detect communication and execution errors.

b7	b6						b0
Error-Flag	Execution error code						

Figure 5: Status byte structure.

The first bit (b7) indicates that at least one of the error flags is set in the Device Status Register.

The “Execution error code” signalizes all errors which occur while processing the frame or executing the command. The following table shows the error codes which can be reported from the device. Note that some of these errors are system internal errors which require additional knowledge to be understood. In case of a problem, they will help Sensirion to localize and solve the issue.

Error Code		Meaning
dec	hex	
0	0x00	No error
1	0x01	Wrong data length for this command (too much or little data)
2	0x02	Unknown command
3	0x03	No access right for command
4	0x04	Illegal command parameter or parameter out of allowed range
40	0x28	Internal function argument out of range
67	0x43	Command not allowed in current state

Table 6 Reference table for error codes.

Data

The data has a usable size of [0…255] bytes (original data, before byte-stuffing). The meaning of the data content depends on the command.

Checksum

The checksum is built before byte-stuffing and checked after removing stuffed bytes from the frame. The checksum is defined as follows:

1. Sum all bytes between start and stop (without start and stop bytes).
2. Take the least significant byte of the result and invert it. This will be the checksum.

For a MOSI frame use Address, Command, Length and Data to calculate the checksum.
For a MISO frame use Address, Command, State, Length and Data to calculate the checksum.

Example (MOSI frame without start/stop and without byte-stuffing):

Adr	CMD	L	Tx Data 2 Bytes	CHK
0x00	0x00	0x02	0x01, 0x03	0xF9

The checksum is calculated as follows:

Adr	0x00
CMD	0x00
L	0x02
Data 0	0x01
Data 1	0x03
Sum	0x06
Least Significant Byte of Sum	0x06
Inverted (=Checksum)	**0xF9**

5.3 **SHDLC Commands**

The following table shows an overview of the available SHDLC commands.

CMD	Command	Read / Write / Execute	max. Response Time	min. required Firmware
0x00	Start Measurement	Execute	20 ms	V1.0
0x01	Stop Measurement	Execute	20 ms	V1.0
0x03	Read Measured Value	Read	20 ms	V1.0
0x10	Sleep	Execute	5 ms	V2.0
0x11	Wake-up	Execute	5 ms	V2.0
0x56	Start Fan Cleaning	Execute	20 ms	V1.0
0x80	Read/Write Auto Cleaning Interval	Read / Write	20 ms	V1.0
0xD0	Device Information	Read	20 ms	V1.0
0xD1	Read Version	Read	20 ms	V1.0
0xD2	Read Device Status Register	Read	20 ms	V2.2
0xD3	Reset	Execute	20 ms	V1.0

Table 7 Reference table for SHDLC commands.

5.3.1 Start Measurement (CMD: 0x00)

Starts the measurement[13]. After power up, the module is in Idle-Mode. Before any measurement values can be read, the Measurement-Mode needs to be started using this command.

MOSI Data:

Byte #	Datatype	Description
0	uint8	Subcommand, this value must be set to **0x01**
1	uint8	Measurement Output Format: **0x03**: Big-endian IEEE754 float values **0x05**: Big-endian unsigned 16-bit integer values

MISO Data: No data.

Example Frames:

MOSI	*Start measurement with output format "Big-endian IEEE754 float values":* **0x7E 0x00 0x00 0x02 0x01 0x03 0xF9 0x7E**
MISO	*Empty response frame:* **0x7E 0x00 0x00 0x00 0x00 0xFF 0x7E**

5.3.2 Stop Measurement (CMD: 0x01)

Stops the measurement[14]. Use this command to return to the initial state (Idle-Mode).

MOSI Data: No data.

MISO Data: No data.

Example Frames:

MOSI	**0x7E 0x00 0x01 0x00 0xFE 0x7E**
MISO	*Empty response frame:* **0x7E 0x00 0x01 0x00 0x00 0xFE 0x7E**

5.3.3 Read Measured Values (CMD: 0x03)

Reads the measured values from the module. This command can be used to poll for new measurement values. The measurement interval is 1 second.

MOSI Data: No data.

MISO Data:

If no new measurement values are available, the module returns an empty response frame.

If new measurement values are available, the response frame contains the measurement results. The data format depends on the selected output format, see 4.3 Measurement Output Formats.

Example Frames:

MOSI	**0x7E 0x00 0x03 0x00 0xFC 0x7E**

[13] This command can only be executed in Idle-Mode.

[14] This command can only be executed in Measurement-Mode.

MISO	*Empty response frame:* **0x7E 0x00 0x03 0x00 0x00 0xFC 0x7E** *Or response frame with new measurement values:* **0x7E 0x00 0x03 0x00 0x28 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00** **0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00** **0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00** **0xD4 0x7E**

5.3.4 Sleep (CMD: 0x10)[15]

Enters the Sleep-Mode with minimum power consumption. This will also deactivate the UART interface, note the wake-up sequence described at the Wake-up command.

MOSI Data: No data.

MISO Data: No data.

Example Frames:

MOSI	**0x7E 0x00 0x10 0x00 0xEF 0x7E**
MISO	**0x7E 0x00 0x10 0x00 0x00 0xEF 0x7E**

5.3.5 Wake-up (CMD: 0x11)

Use this command to switch from Sleep-Mode to Idle-Mode. In Sleep-Mode the UART interface is disabled and must first be activated by sending a low pulse on the RX pin. This pulse is generated by sending a single byte with the value 0xFF.

If then a Wake-up command follows within 100ms, the module will switch on again and is ready for further commands in the Idle-Mode. If the low pulse is not followed by the Wake-up command, the microcontroller returns to Sleep-Mode after 100ms and the interface is deactivated again.

The Wake-up command can be sent directly after the 0xFF, without any delay. However, it is important that no other value than 0xFF is used to generate the low pulse, otherwise it's not guaranteed the UART interface synchronize correctly.

MOSI Data: No data.

MISO Data: No data.

Example Frames:

MOSI	*Send 0xFF to generate a low pulse in order to wake-up the interface:* **0xFF** *Wake-up command, within 100ms:* **0x7E 0x00 0x11 0x00 0xEE 0x7E**
MISO	**0x7E 0x00 0x11 0x00 0x00 0xEE 0x7E**

Alternatively, if the software implementation does not allow to send a single byte with the value 0xFF, the Wake-up command can be sent twice in succession. In this case the first Wake-up command is ignored, but causes the interface to be activated.

[15] This command can only be executed in Idle-Mode.

MOSI	*First Wake-up command (just, activates the interface):* **0x7E 0x00 0x11 0x00 0xEE 0x7E** *Second Wake-up command, within 100ms (this finally wakes up the module):* **0x7E 0x00 0x11 0x00 0xEE 0x7E**
MISO	**0x7E 0x00 0x11 0x00 0x00 0xEE 0x7E**

5.3.6 Start Fan Cleaning (CMD: 0x56)

Starts the fan-cleaning manually[16]. For more details, note the explanations given in 4.2 Fan Auto Cleaning.

MOSI Data: No data.

MISO Data: No data.

Example Frames:

MOSI	**0x7E 0x00 0x56 0x00 0xA9 0x7E**
MISO	**0x7E 0x00 0x56 0x00 0x00 0xA9 0x7E**

5.3.7 Read/Write Auto Cleaning Interval (CMD: 0x80)

Reads/Writes the interval [s] of the periodic fan-cleaning. For more details, note the explanations given in 4.2 Fan Auto Cleaning.

MOSI Data:

Read Auto Cleaning Interval:

Byte #	Datatype	Description
0	uint8	Subcommand, this value must be set to **0x00**

Write Auto Cleaning Interval:

Byte #	Datatype	Description
0	uint8	Subcommand, this value must be set to **0x00**
1..4	uint32	Interval in seconds as big-endian unsigned 32-bit integer value.

MISO Data:

Read Auto Cleaning Interval:

Byte #	Datatype	Description
0..3	uint32	Interval in seconds as big-endian unsigned 32-bit integer value.

Write Auto Cleaning Interval: No data.

Example Frames:

MOSI	*Read Auto Cleaning Interval:* **0x7E 0x00 0x80 0x01 0x00 0x7D 0x5E 0x7E** *Write Auto Cleaning Interval to 0 (disable):* **0x7E 0x00 0x80 0x05 0x00 0x00 0x00 0x00 0x00 0x7A 0x7E**
MISO	*Response frame for "Read Auto Cleaning Interval":* **0x7E 0x00 0x80 0x00 0x04 0x00 0x00 0x00 0x00 0x7B 0x7E** *Response frame for "Write Auto Cleaning Interval":* **0x7E 0x00 0x80 0x00 0x00 0x7F 0x7E**

[16] This command can only be executed in Measurement-Mode.

5.3.8 Device Information (CMD 0xD0)

This command returns the requested device information. It is defined as a string value with a maximum length of 32 ASCII characters (including terminating null character).

MOSI Data:

Byte #	Datatype	Description
0	uint8	This parameter defines which information is requested: **0x00**: Product Type **0x01**: Reserved **0x02**: Reserved **0x03**: Serial Number

MISO Data:

Byte #	Datatype	Description
0…n	string	Requested Device Information as null-terminated ASCII string. The size of the string is limited to 32 ASCII characters (including null character).

Example Frames:

Product Type:

Recommended to use as product identifier, returns always the string “00080000” on this product.

MOSI	**0x7E 0x00 0xD0 0x01 0x00 0x2E 0x7E**
MISO	**0x7E 0x00 0xD0 0x00 0x09 0x30 0x30 0x30 0x38 0x30 0x30 0x30 0x30 0x00 0x9B 0x7E**

Serial Number:

MOSI	**0x7E 0x00 0xD0 0x01 0x03 0x2B 0x7E**
MISO	**0x7E 0x00 0xD0 0x00 0x15 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x00 0x5A 0x7E**

5.3.9 Read Version (0xD1)

Gets version information about the firmware, hardware, and SHDLC protocol.

MOSI Data: No data.

MISO Data:

Byte #	Datatype	Description
0	uint8	Firmware major version
1	uint8	Firmware minor version[17]
2	uint8	Reserved: always 0
3	uint8	Hardware revision
4	uint8	Reserved: always 0
5	uint8	SHDLC protocol major version
6	uint8	SHDLC protocol minor version

[17] Firmware minor version may change without notice, given full backwards compatibility.

Example Frame:

MOSI	**0x7E 0x00 0xD1 0x00 0x2E 0x7E**
MISO	*Firmware V2.1, Hardware V6, SHDLC V2.0:* **0x7E 0x00 0xD1 0x01 0x07 0x02 0x01 0x00 0x06 0x00 0x02 0x00 0x1C 0x7E**

5.3.10 Read Device Status Register (0xD2)

Use this command to read the Device Status Register. For more details, note the explanations given in 4.4 Device Status Register.

Note: If one of the device status flags of type "Error" is set, this is also indicated in every SHDLC response frame by the Error-Flag in the state byte.

MOSI Data:

Byte #	Datatype	Description
0	uint8	0: Do not clear any bit in the Device Status Register after reading. 1: Clear all bits in the Device Status Register after reading.

MISO Data:

Byte #	Datatype	Description
0…3	big-endian, uint32	Device Status Register
4	uint8	Reserved for future use

Example Frame:

MOSI	**0x7E 0x00 0xD2 0x01 0x00 0x2C 0x7E**
MISO	**0x7E 0x00 0xD2 0x00 0x05 0x00 0x00 0x00 0x00 0x00 0x28 0x7E**

5.3.11 Device Reset (CMD: 0xD3)

Soft reset command. After calling this command, the module is in the same state as after a Power-Reset. The reset is executed after sending the MISO response frame.

Note: To perform a reset when the sensor is in sleep mode, it is required to send first a wake-up sequence to activate the interface.

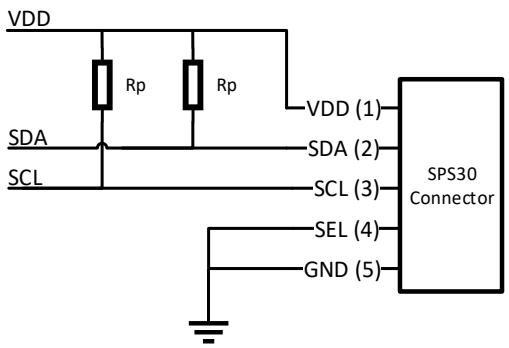
MOSI Data: No data.

MISO Data: No data.

Example Frames:

MOSI	**0x7E 0x00 0xD3 0x00 0x2C 0x7E**
MISO	**0x7E 0x00 0xD3 0x00 0x00 0x2C 0x7E**

6 Operation and Communication through the I²C Interface



Usage:

- I²C address: 0x69
- Max. speed: standard mode, 100 kbit/s
- Clock stretching: not used

Both SCL and SDA lines are open drain I/Os. They should be connected to external pull-up resistors (e.g. Rp = 10 kΩ). **Important notice:** in order to correctly select I²C as interface, the interface select (SEL) pin must be pulled to GND before or at the same time the sensor is powered up.

Figure 6: Typical I²C application circuit.

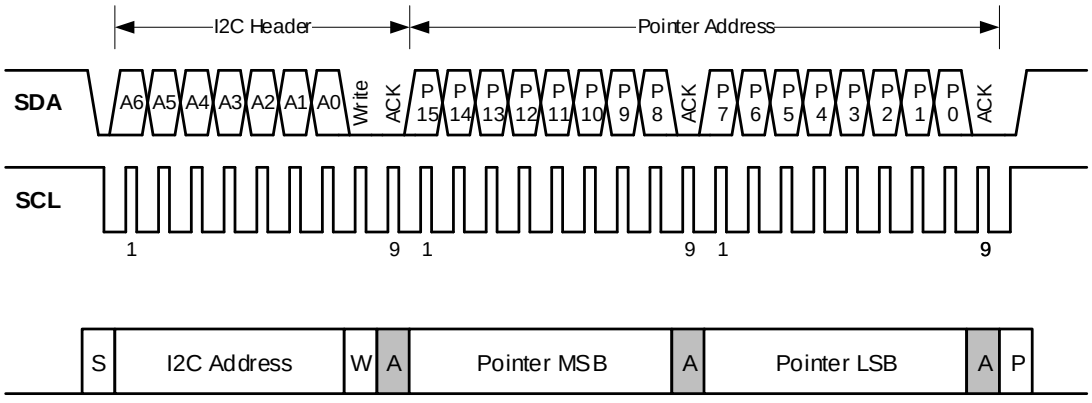
Some considerations should be made about the use of the I²C interface. I²C was originally designed to connect two chips on a PCB. When the sensor is connected to the main PCB via a cable, particular attention must be paid to electromagnetic interference and crosstalk. Use as short as possible (< 10 cm) and/or well shielded connection cables. We recommend using the UART interface instead, whenever possible: it is more robust against electromagnetic interference, especially with long connection cables.

For detailed information on the I2C protocol, refer to NXP I2C-bus specification[18].

6.1 Transfer Types

Set Pointer

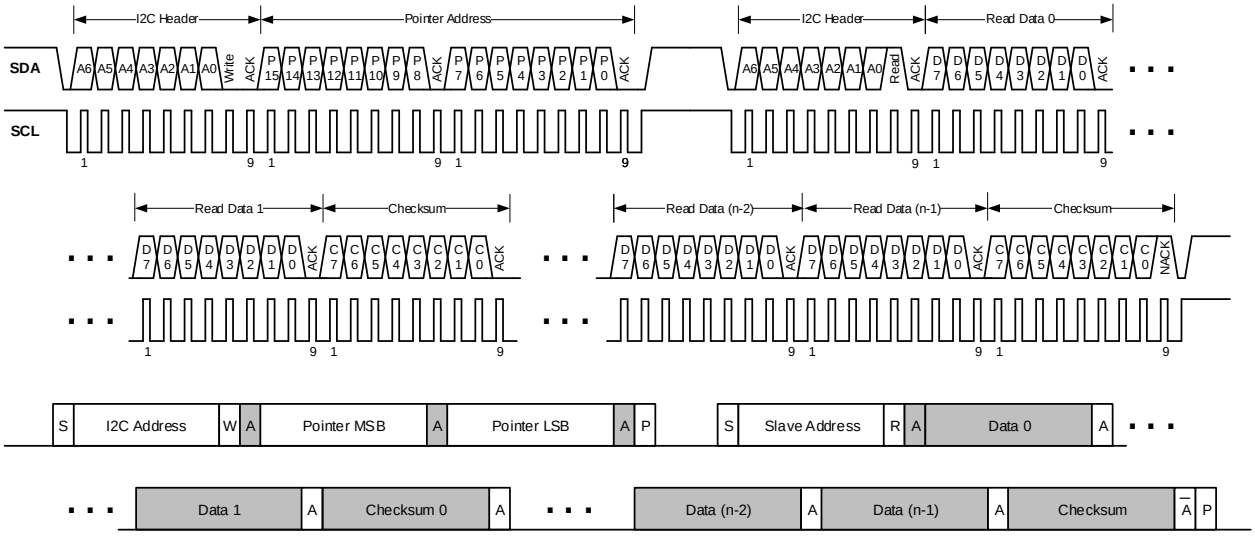
Sets the 16-bit address pointer without writing data to the sensor module. It is used to execute commands, which do not require additional parameters.



[18] http://www.nxp.com/documents/user_manual/UM10204.pdf

Set Pointer & Read Data

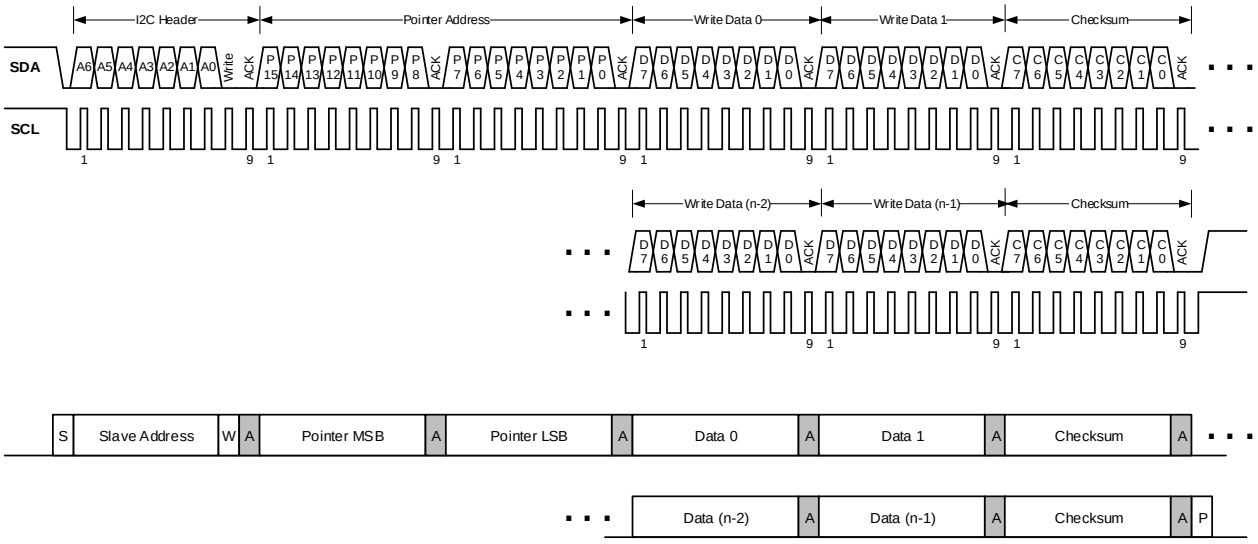
Sets the 16-bit address pointer and read data from sensor module. It is used to read sensor module information or measurement results. The data is ready to read immediately after the address pointer is set. The sensor module transmits the data in 2-byte packets, which are protected with a checksum.



It is allowed to read several times in succession without setting the address pointer again. This reduces the protocol overhead for periodical reading of the measured values.

Set Pointer & Write Data

Sets the 16-bit address pointer and writes data to the sensor module. It is used to execute commands, which require additional parameters. The data must be transmitted in 2-byte packets which are protected by a checksum.



6.2 Checksum Calculation

The Read and Write Commands transmit the data in 2-byte packets, followed by an 8-bit checksum. The checksum is calculated as follows:

Property	Value
Name	CRC-8
Protected Data	read and/or write data
Width	8 bit
Polynomial	0x31 (x^8 + x^5 + x^4 + 1)
Initialization	0xFF
Reflect Input	false
Reflect Output	false
Final XOR	0x00
Example	CRC(0xBEEF) = 0x92

```
uint8_t CalcCrc(uint8_t data[2]) {
  uint8_t crc = 0xFF;
  for(int i = 0; i < 2; i++) {
    crc ^= data[i];
    for(uint8_t bit = 8; bit > 0; --bit) {
      if(crc & 0x80) {
        crc = (crc << 1) ^ 0x31u;
      } else {
        crc = (crc << 1);
      }
    }
  }
  return crc;
}
```

Please note that the checksums are used only for the 2-byte data packets. The command code itself already contains a 3-bit CRC and therefore no checksum must be appended to it.

6.3 I2C Commands

The following table shows an overview of the available I²C commands.

Address Pointer	Command Name	Transfer Type	Parameter length including CRC [bytes]	Response length including CRC [bytes]	Command execution time	min. required Firmware
0x0010	Start Measurement	Set Pointer & Write Data	3	-	< 20 ms	V1.0
0x0104	Stop Measurement	Set Pointer	-	-	< 20 ms	V1.0
0x0202	Read Data-Ready Flag	Set Pointer & Read Data	-	3	-	V1.0
0x0300	Read Measured Values	Set Pointer & Read Data	-	float: 60 integer: 30	-	V1.0
0x1001	Sleep	Set Pointer	-	-	< 5 ms	V2.0
0x1103	Wake-up	Set Pointer	-	-	< 5 ms	V2.0
0x5607	Start Fan Cleaning	Set Pointer	-	-	< 5 ms	V1.0
0x8004	Read/Write Auto Cleaning Interval	Set Pointer & Read/Write Data	read: - write: 6	read: 6 write: -	read: < 5 ms write: < 20 ms	V1.0
					read: - write: < 20ms	V2.2
0xD002	Read Product Type	Set Pointer & Read Data	-	12		
0xD033	Read Serial Number	Set Pointer & Read Data	-	max. 48	-	V1.0
0xD100	Read Version	Set Pointer & Read Data	-	3	-	V1.0
0xD206	Read Device Status Register	Set Pointer & Read Data	-	6	-	V2.2
0xD210	Clear Device Status Register	Set Pointer	-	-	< 5 ms	V2.0
0xD304	Reset	Set Pointer	-	-	< 100 ms	V1.0

Table 8 Reference table for I²C commands.

6.3.1 Start Measurement (0x0010)

Starts the measurement. After power up, the module is in Idle-Mode. Before any measurement values can be read, the Measurement-Mode needs to be started using this command.

Transfer Type: **Set Pointer & Write Data**
Pointer Address: **0x0010**

Write Data:

Byte #	Description
0	**Measurement Output Format** **0x03**: Big-endian IEEE754 float values **0x05**: Big-endian unsigned 16-bit integer values
1	dummy byte, insert 0x00
2	Checksum for bytes 0, 1

6.3.2 Stop Measurement (0x0104)

Stops the measurement. Use this command to return to the Idle-Mode.

Transfer Type: **Set Pointer**
Pointer Address: **0x0104**

6.3.3 Read Data-Ready Flag (0x0202)

This command can be used for polling to find out when new measurements are available. The pointer address only has to be set once. Repeated read requests get the status of the Data-Ready Flag.

Transfer Type: **Set Pointer & Read Data**
Pointer Address: **0x0202**

Read Data:

Byte #	Description
0	unused, always 0x00
1	**Data-Ready Flag** 0x00: no new measurements available 0x01: new measurements ready to read
2	Checksum for bytes 0, 1

6.3.4 Read Measured Values (0x0300)

Reads the measured values from the sensor module and resets the “Data-Ready Flag”. If the sensor module is in Measurement-Mode, an updated measurement value is provided every second and the “Data-Ready Flag” is set. If no synchronized readout is desired, the “Data-Ready Flag” can be ignored. The command “Read Measured Values” always returns the latest measured values.

Transfer Type: **Set Pointer & Read Data**
Pointer Address: **0x0300**

The data format depends on the selected output format, see 4.3 Measurement Output Formats. Note that after every two bytes, the checksum of the previous two bytes is transferred.

Example Data Structure:

Byte #	Description
0, 1	two bytes of measurement data
2	checksum for bytes 0, 1
3, 4	two bytes of measurement data
5	checksum for bytes 3, 4
…	…

6.3.5 Sleep (0x1001)[19]

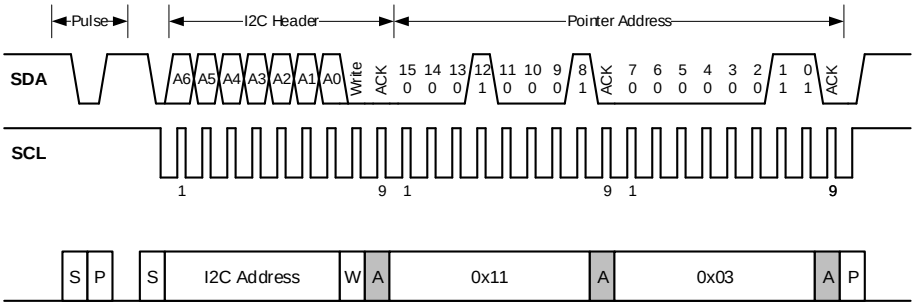
Enters the Sleep-Mode with minimum power consumption. This will also deactivate the I2C interface, note the wake-up sequence described at the Wake-up command.

Transfer Type: **Set Pointer**
Pointer Address: **0x1001**

6.3.6 Wake-Up (0x1103)

Use this command to switch from Sleep-Mode to Idle-Mode. In Sleep-Mode the I2C interface is disabled and must first be activated by sending a low pulse on the SDA line. A low pulse can be generated by sending a I2C-Start-Condition followed by a Stop-Condition.

If then a Wake-up command follows within 100ms, the module will switch on again and is ready for further commands in the Idle-Mode. If the low pulse is not followed by the Wake-up command, the microcontroller returns after 100ms to Sleep-Mode and the interface is deactivated again.



Alternatively, if the software implementation does not allow to send a I2C-Start-Condition followed by a Stop-Condition, the Wake-up command can be sent twice in succession. In this case the first Wake-up command is ignored, but causes the interface to be activated.

Transfer Type: **2x Set Pointer**
Pointer Address: **0x1103**

6.3.7 Start Fan Cleaning (0x5607)[20]

Starts the fan-cleaning manually. For more details, note the explanations given in 4.2 Fan Auto Cleaning.

Transfer Type: **Set Pointer**
Pointer Address: **0x5607**

[19] This command can only be executed in Idle-Mode.
[20] This command can only be executed in Measurement-Mode.

6.3.8 Read/Write Auto Cleaning Interval (0x8004)

Reads/Writes the interval [s] of the periodic fan-cleaning. For more details, note the explanations given in 4.2 Fan Auto Cleaning.

Note for FW Version < 2.2: After writing a new interval, this will be activated immediately. However, if the interval register is read out after setting the new value, the previous value is returned until the next start/reset of the sensor module.

Transfer Type: **Set Pointer & Read/Write Data**
Pointer Address: **0x8004**

Read/Write Data:

Byte #	Description	
0, 1	Most Significant Byte	big-endian, unsigned 32-bit integer value: **Auto Cleaning Interval [s]**
2	Checksum for bytes 0, 1	
3, 4	Least Significant Byte	
5	Checksum for bytes 3, 4	

6.3.9 Read Device Information (0xD002, 0xD033)

This command returns the requested device information. It is defined as a string value with a maximum length of 32 ASCII characters (including terminating null-character).

Transfer Type: **Set Pointer & Read Data**
Pointer Address: **Product Type:** **0xD002** (always "00080000" without terminating null-character, recommended to use as product identifier)
Serial Number: **0xD033**

Read Data:

Byte #	Description
0	ASCII Character 0
1	ASCII Character 1
2	Checksum for bytes 0, 1
…	…
45	ASCII Character 30
46	ASCII Character 31
47	Checksum for bytes 45, 46

6.3.10 Read Firmware Version (0xD100)

Gets firmware major.minor version.

Transfer Type: **Set Pointer & Read Data**
Pointer Address: **0xD100**

Read Data:

Byte #	Description
0	Firmware major version
1	Firmware minor version[21]
2	Checksum for bytes 0, 1

[21] Firmware minor version may change without notice, given full backwards compatibility.

6.3.11 Read Device Status Register (0xD206)

Use this command to read the Device Status Register. For more details, note the explanations given in 4.4.

Transfer Type: **Set Pointer & Read Data**
Pointer Address: **0xD206**

Read Data:

Byte #	Description	
0, 1	Most Significant Byte	big-endian, unsigned 32-bit integer value: **Device Status Register**
2	Checksum for bytes 0, 1	
3, 4	Least Significant Byte	
5	Checksum for bytes 3, 4	

6.3.12 Clear Device Status Register (0xD210)

Clears the device status register. For more details, note the explanations given in 4.4.

Transfer Type: **Set Pointer**
Pointer Address: **0xD210**

6.3.13 Device Reset (0xD304)

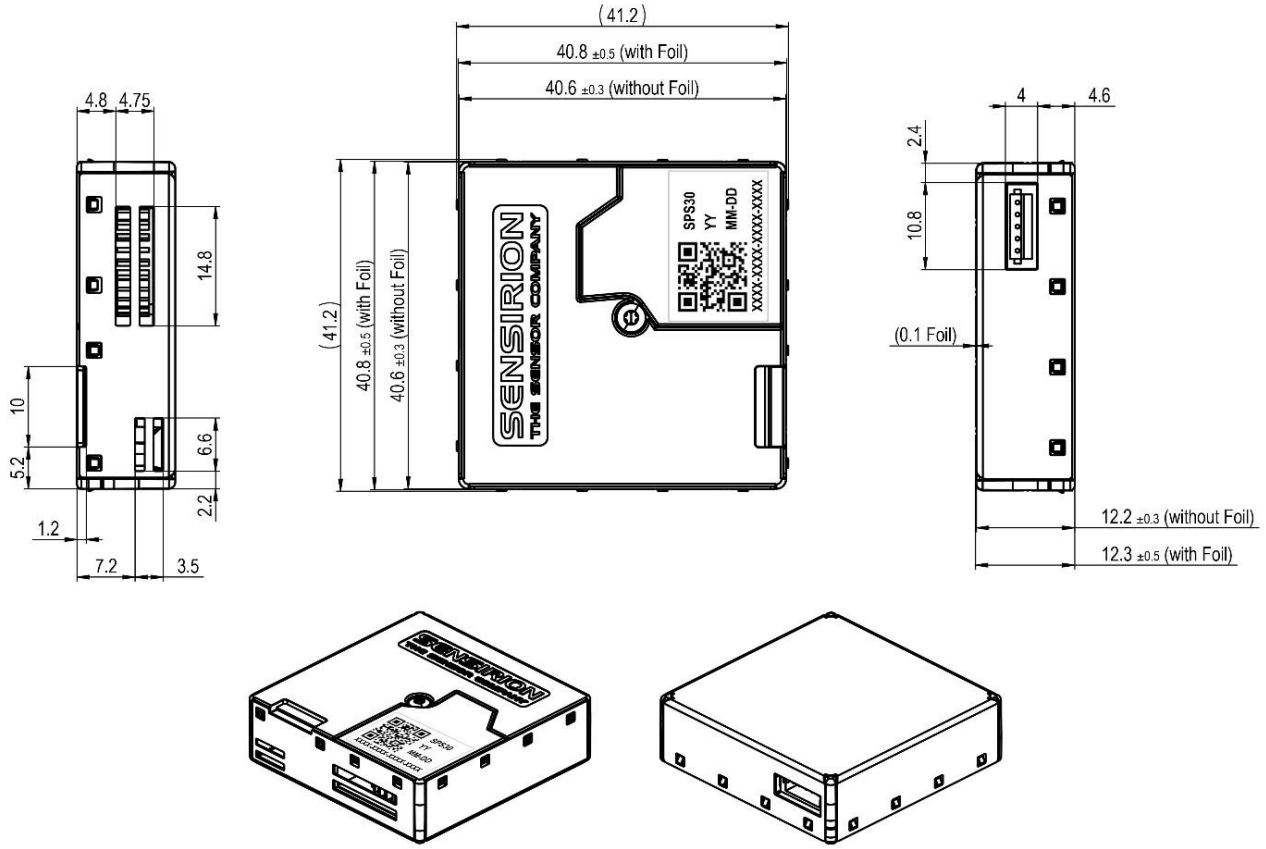
Device software reset command. After calling this command, the module is in the same state as after a power reset.

Note: To perform a reset when the sensor is in sleep mode, it is required to send first a wake-up sequence to activate the interface.

Transfer Type: **Set Pointer**
Pointer Address: **0xD304**

7 Mechanical Specifications

7.1 Product Outline Drawings



General tolerances: ISO 2768-mK

Figure 7: Package outline dimensions of the SPS30 from different views. Tolerances included. All lengths are given in mm. Dimensions in brackets include plastic fixation elements.

7.2 Label

The SPS30 has a label on one side of the sensor. The label contains information about

- Sensirion product name
- Production year
- Production date (month - day)
- Serial number (XXXX-XXXX-XXXX-XXXX)
- QR-code containing the information above



Figure 8: Label on SPS30

8 Shipping Package

The SPS30 is shipped in stackable trays with 56 pieces each. Non-packaged tray dimensions are given in Figure 9. Packaged tray dimensions are 670 mm x 460 mm x 45 mm. The weight of each full packaged tray (including sensors) is 2.4 kg.

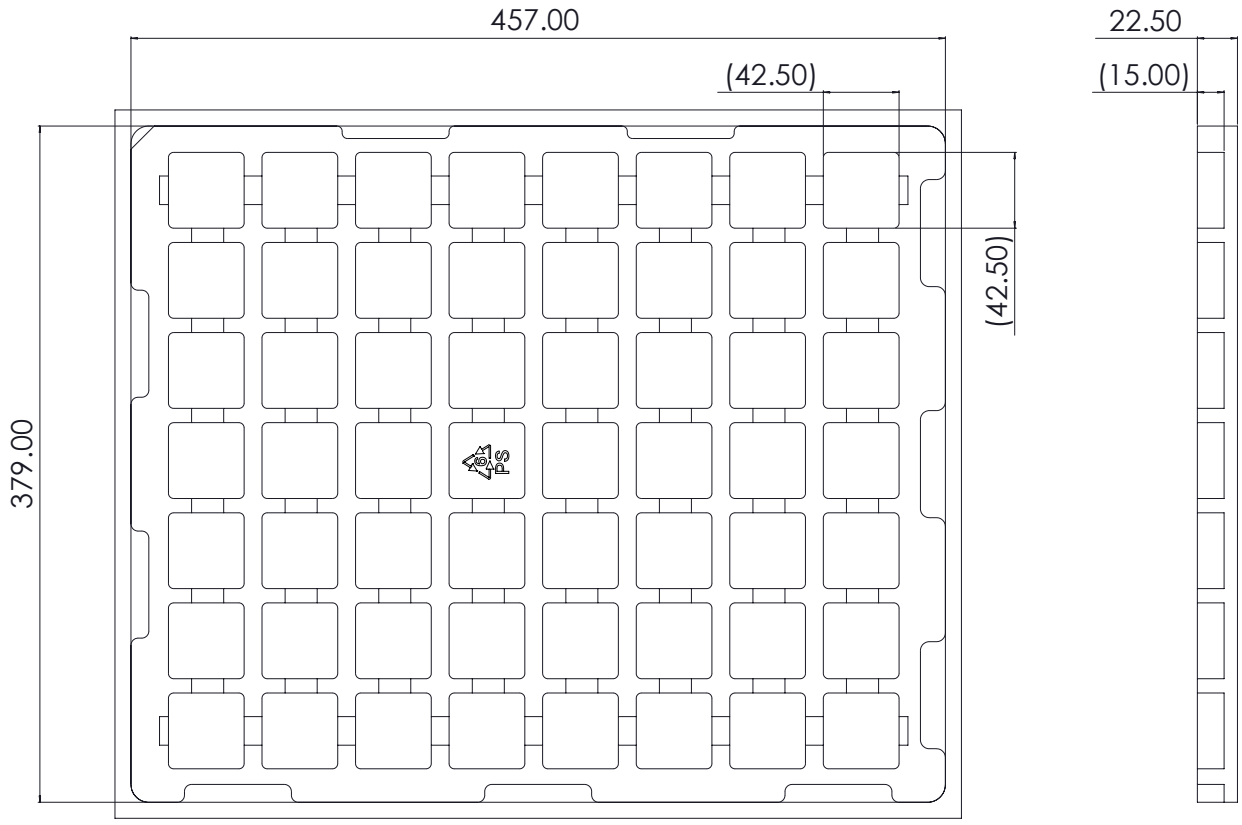


Figure 9: 56-sensor tray dimensions (in mm).

9 Ordering Information

The SPS30 and its evaluation kit can be ordered via the article numbers listed in Table 9.

Product	Description	Article Number
SPS30 sensor	Particulate Matter Sensor	1-101638-10
SEK-SPS30 evaluation kit	SPS30 sensor and USB evaluation kit	3.000.119

Table 9: SPS30 and evaluation kit ordering information.

10 Revision History

Date	Version	Page(s)	Changes
27. March 2020	1.0	All	Initial version
27. June 2023	2.0	2	Improved mass concentration precision specification for PM1 and PM2.5 (Table 1)

Important Notices

Warning, Personal Injury

Do not use this product as safety or emergency stop devices or in any other application where failure of the product could result in personal injury. Do not use this product for applications other than its intended and authorized use. Before installing, handling, using or servicing this product, please consult the data sheet and application notes. Failure to comply with these instructions could result in death or serious injury.

If the Buyer shall purchase or use SENSIRION products for any unintended or unauthorized application, Buyer shall defend, indemnify and hold harmless SENSIRION and its officers, employees, subsidiaries, affiliates and distributors against all claims, costs, damages and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if SENSIRION shall be allegedly negligent with respect to the design or the manufacture of the product.

ESD Precautions

The inherent design of this component causes it to be sensitive to electrostatic discharge (ESD). To prevent ESD-induced damage and/or degradation, take customary and statutory ESD precautions when handling this product. See application note "ESD, Latchup and EMC" for more information.

Warranty

SENSIRION warrants solely to the original purchaser of this product for a period of 12 months (one year) from the date of delivery that this product shall be of the quality, material and workmanship defined in SENSIRION's published specifications of the product. Within such period, if proven to be defective, SENSIRION shall repair and/or replace this product, in SENSIRION's discretion, free of charge to the Buyer, provided that:

- notice in writing describing the defects shall be given to SENSIRION within fourteen (14) days after their appearance;
- such defects shall be found, to SENSIRION's reasonable satisfaction, to have arisen from SENSIRION's faulty design, material, or workmanship;
- the defective product shall be returned to SENSIRION's factory at the Buyer's expense; and
- the warranty period for any repaired or replaced product shall be limited to the unexpired portion of the original period.

This warranty does not apply to any equipment which has not been installed and used within the specifications recommended by SENSIRION for the intended and proper use of the equipment. EXCEPT FOR THE WARRANTIES EXPRESSLY SET FORTH HEREIN, SENSIRION MAKES NO WARRANTIES, EITHER EXPRESS OR IMPLIED, WITH RESPECT TO THE PRODUCT. ANY AND ALL WARRANTIES, INCLUDING WITHOUT LIMITATION, WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, ARE EXPRESSLY EXCLUDED AND DECLINED.

SENSIRION is only liable for defects of this product arising under the conditions of operation provided for in the data sheet and proper use of the goods. SENSIRION explicitly disclaims all warranties, express or implied, for any period during which the goods are operated or stored not in accordance with the technical specifications.

SENSIRION does not assume any liability arising out of any application or use of any product or circuit and specifically disclaims any and all liability, including without limitation consequential or incidental damages. All operating parameters, including without limitation recommended parameters, must be validated for each customer's applications by customer's technical experts. Recommended parameters can and do vary in different applications.

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www.sensirion.com/cn

To find your local representative, please visit www.sensirion.com/distributors

RK400-04 Economical Tipping Bucket Rainfall Sensor

www.rikasensor.com.cn

RK400-04 Economical Tipping Bucket Rainfall Sensor is a hydrology, meteorological instrument used to measure the nature of rainfall, and it converts the precipitation into a pulse signal output. Designed with insect-proof nets, free blocking nozzle and built-in leveller. The model can be used in meteorology, hydrology, agriculture, forestry, field monitoring stations and other industries .Combined with rainfall recorder can be used to measure measure precipitation, precipitation intensity, precipitation time.

FEATURES

- Compact size for easy use
- High accuracy, good stability
- Mesh in the funnel preventing debris such as leaves and insects from entering the working of rain sensor
- Well made tipping bucket with low resistance
- The main body made of high strength ABS
- Horizontal Bubble in the bottom
- Outlet with insect-proof screen



APPLICATIONS

- Meteorological monitoring
- Hydrologic monitoring
- Natural disaster monitoring
- Agrometeorological research
- Climate research

SPECIFICATIONS

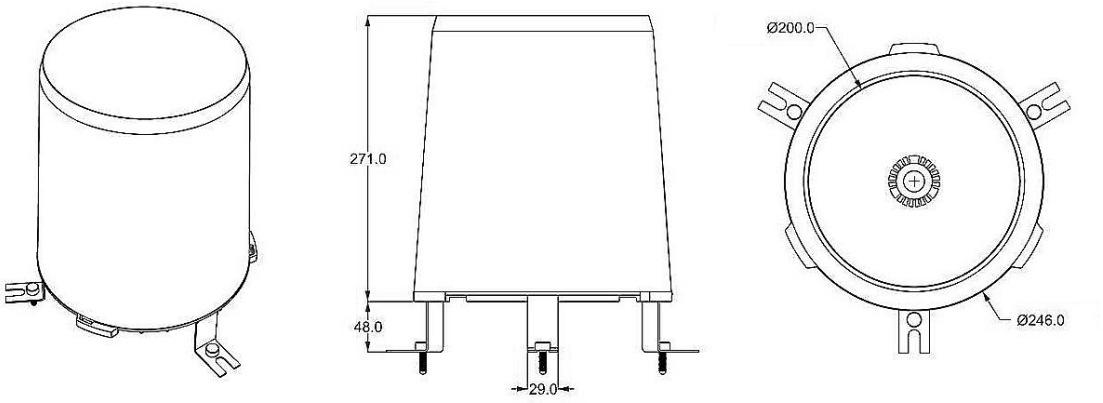
Item	Specification
Collector	Diameter :φ200mm, height: 271mm
Measured rainfall intensity	Max: 4mm/min
Allow rainfall intensity	Max: 8mm/min
Resolution	0.2mm
Accuracy(2mm/min)	±4%
Maximum load voltage	30VDC(pulse output)
Maximum load current	20mA
Output	Pulses(@10kΩ&0.01uF),RS485(12-24VDC supply)
Operating temperature	0-60℃@0%-100%RH
Main material	Rainfall collector & tipping bucket: ABS, Supporting leg:304SS
Weight(unpacked)	2kg

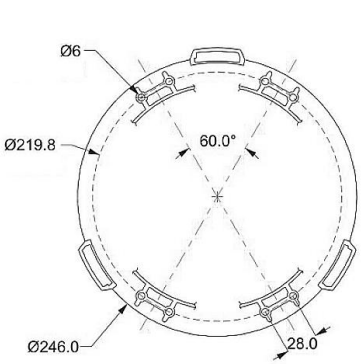
WORKING PROCESS

Rainfall is captured in the 200mm diameter collector funnel and is directed through a delivery pipe to fill a divided ABS injection molded tipping bucket device.The bucket is pivoted through its center and has a preset calibration to tip for 0.2 mm of rainfall. When the bucket is “full”, it pivots and empties - this action magnetically closes and opens a reed switch, sending a pulse signal to the data logger or electronic counter. Through this tipping “seesaw” action, the other side of the bucket is aligned to receive the flow from the delivery pipe. This recording and tipping cycle continues with rainfall.

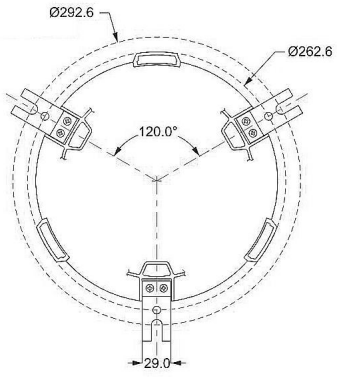
DIMENSION & MOUNTING

Adjust the balance foot screws to ensure that it is in a horizontal position.Using three fastening screws to fix this three support legs at the platform to keep the entire apparatus at an optimum level , which ensure the accuracy of rainfall data.





Installation hole position No.1



Installation hole position No.2

PARAMETER SELECTION TABLE

Remark	Series	Type	Output	Cable Length	
RK					
	400				
		04			
			A		Pulse
			B		RS485(supply:12-24VDC)
			C		Customization
				1500	Units:mm（typ）
				3000	Units:mm
				...	Units:mm

Example: RK400-04A1500 Output:pulse Cable Length:1.5m.

CE Complies with applicable CE directives.
Specifications subject to change without notice. Version 3.0
Copyright © 2015 Hunan Rika Electronic Technology Co.,Ltd

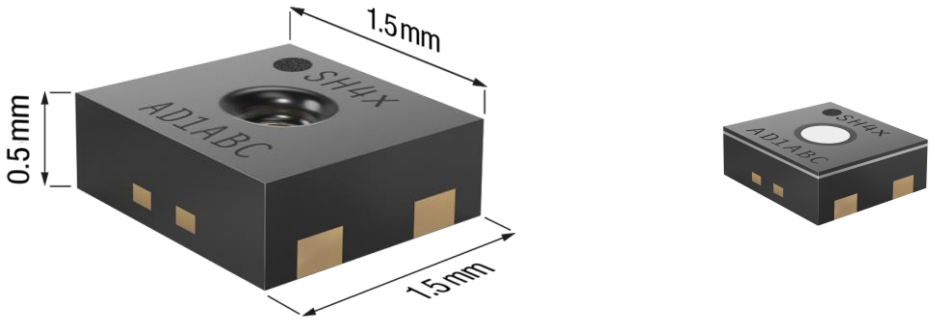
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Yuhua District,Changsha,
Hunan,China

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info@rikasensor.com
www.rikasensor.com.cn

SENSIRION

SHT4x

4th Generation, High-Accuracy, Ultra-Low-Power, 16-bit Relative Humidity and Temperature Sensor



Features

- Accuracies Δ*RH* = ±1.0 %RH, Δ*T* = ±0.1 °C
- VDD = 1.08 V … 3.6 V
- Avg. current: 0.4 µA, Idle current: 80 nA
- I2C FM+, CRC checksum, multip. I2C addr.
- Patented protection options [1], PTFE membrane and removable protective cover
- Operating range: 0 … 100 %RH, −40…125 °C
- Fully functional in condensing environment
- Power heater, true NIST-traceability
- JEDEC JESD47 qualification
- Sensor-specific calibration certificate acc. to ISO 17025 : 2017, 3-point temp. calibration

General Description

SHT4x is a digital sensor platform for measuring relative humidity and temperature at different accuracy classes. Its I2C interface provides several preconfigured I2C addresses while maintaining an ultra-low power budget. The power-trimmed internal heater can be used at three heating levels thus enabling sensor operation in demanding environments. The four-pin dual-flat-no-leads package is suitable for surface mount technology (SMT) processing and comprises an optional on-package patented PTFE [1] membrane or a removable protective cover. Sensor specific calibration certificates according to ISO17025, identifiable through unique serial number, are available.

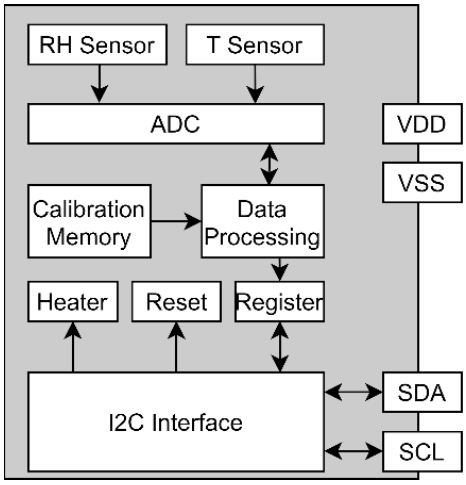
Device Overview

Products	Details
SHT40-AD1B	base RH&T accur., 0x44 I2C addr.
SHT40-BD1B	base RH&T accur., 0x45 I2C addr.
SHT40-AD1F	SHT40-AD1B with PTFE membrane
SHT40-AD1P	SHT40-AD1B with protective cover
SHT41-AD1B	intermed. RH&T accur., 0x44 I2C addr.
SHT43-ADCB	ISO17025 3-point calibration certificate
SHT45-AD1B	±1.0 %RH, ±0.1 °C accur., 0x44 I2C addr.

Full product list on page 19

SCAN ME to give feedback and help us to improve this document. —Thank you!

Functional Block Diagram



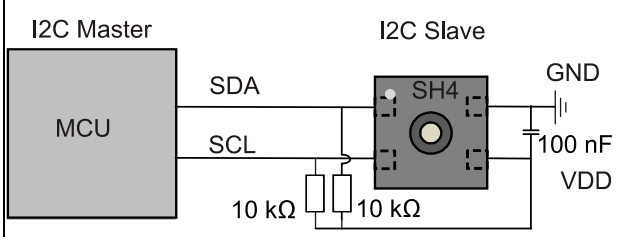
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1 Quick Start – Hello World

A typical application circuit for SHT4x is shown on the left-hand side of **Figure 1**. After reaching the minimal supply voltage and allowing for the maximal power-up time of 1 ms the sensor is ready for I2C communication. The quickest way to measure humidity and temperature is pseudo-coded on the right-hand side of **Figure 1**. Together with the conversion formulae given in equations (1), (2), and (3), the digital signals can be translated into relative humidity and temperature readings.

Typical application circuit



Pseudo code

```
i2c_write(i2c_addr=0x44, tx_bytes=[0xFD])
wait_seconds(0.01)
rx_bytes = i2c_read(i2c_addr=0x44,
number_of_bytes=6)
t_ticks = rx_bytes[0] * 256 + rx_bytes[1]
checksum_t = rx_bytes[2]
rh_ticks = rx_bytes[3] * 256 + rx_bytes[4]
checksum_rh = rx_bytes[5]
t_degC = -45 + 175 * t_ticks/65535
rh_pRH = -6 + 125 * rh_ticks/65535
if (rh_pRH > 100):
      rh_pRH = 100
if (rh_pRH < 0):
      rh_pRH = 0
```

Figure 1. Typical application circuit (left) and pseudo code (right) for easy starting. For details on the signal cropping in the last four lines see section 4.6.

Find code resources and embedded drivers on: https://github.com/Sensirion/embedded-sht/releases

CAD files can be downloaded from SnapEDA: https://www.snapeda.com/parts/SHT40-AD1B-R3/Sensirion/view-part/

2 Humidity and Temperature Sensor Specifications

Every SHT4x is individually tested and calibrated and is identifiable by its unique serial number (see section 4.7 for details on the serial number). For the calibration, Sensirion uses transfer standards, which are subject to a scheduled calibration procedure. The calibration of the reference, used for the calibration of the transfer standards, is NIST traceable through an ISO/IEC 17025 accredited laboratory.

2.1 Relative Humidity

Parameter	Conditions	Value	Units
SHT40 *RH* accuracy[1]	typ.	±1.8	%RH
	max.	see **Figure 2**	-
SHT41 *RH* accuracy[1]	typ.	±1.8	%RH
	max.	see **Figure 3**	-
SHT43 *RH* accuracy[1]	typ.	±1.8	%RH
	max.	see **Figure 4**	-
SHT45 *RH* accuracy[1]	typ.	±1.0	%RH
	max.	see **Figure 5**	-
Repeatability[2, 3]	high	0.08	%RH
	medium	0.15	%RH
	low	0.25	%RH
Resolution[4]	-	0.01	%RH
Hysteresis	At 25 °C	±0.8	%RH
Specified range[5]	extended[6]	0 to 100	%RH
Response time[7]	$\tau_{63\%}$	4	s
Long-term drift[8]	typ.	<0.2	%RH/y

Table 1. General relative humidity sensor specifications.

[1] For definition of typ. and max. accuracy, please refer to the document "Sensirion Humidity Sensor Specification Statement".

[2] The stated repeatability is three times the standard deviation (3σ) of multiple consecutive measurement values at constant conditions and is a measure for the noise on the physical sensor output. Different repeatability commands are listed in **Table 8**.

[3] Valid for 25 °C and 50 %RH.

[4] Resolution of A/D converter.

[5] Specified range refers to the range for which the humidity or temperature sensor specification is guaranteed.

[6] For details about recommended humidity and temperature operating range, please refer to section 2.3.

[7] Time for achieving 63% of a humidity step function, measured at 25 °C and 1 m/s airflow. Humidity response time in the application depends on the design-in of the sensor.

[8] Typical value for operation in normal RH/T operating range. Value may be higher in environments with vaporized solvents, out-gassing tapes, adhesives, packaging materials, etc. For more details please refer to Handling Instructions.

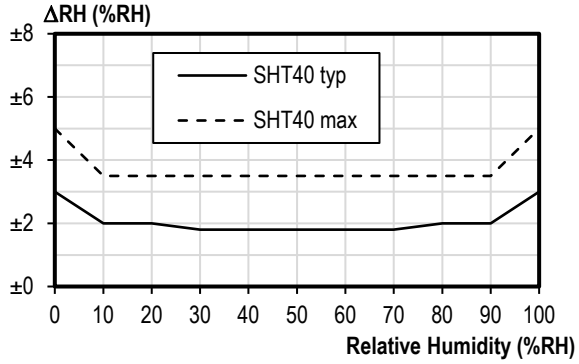


Figure 2. SHT40 typical and maximal relative humidity accuracy at 25 °C.

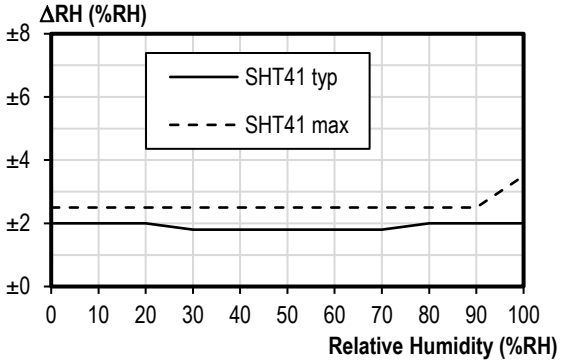


Figure 3. SHT41 typical and maximal relative humidity accuracy at 25 °C.

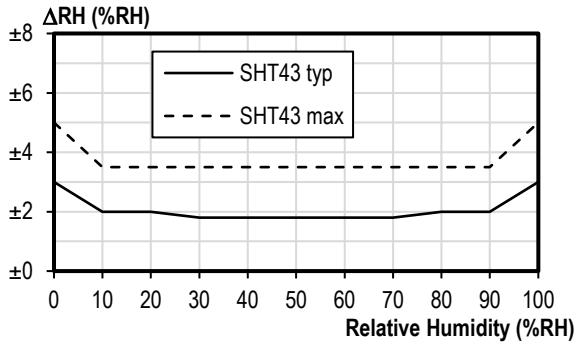


Figure 4. SHT43 typical and maximal relative humidity accuracy at 25 °C.

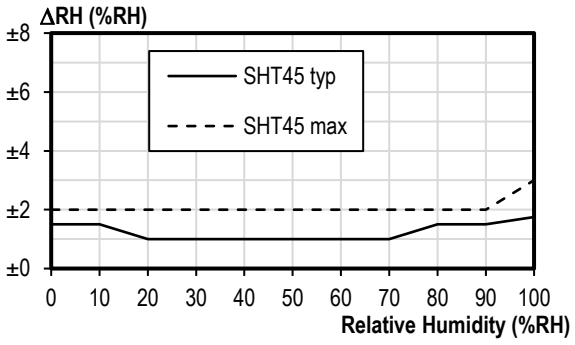


Figure 5. SHT45 typical and maximal relative humidity accuracy at 25 °C.

Relative Humidity Accuracy at the Extended Temperature Range

The typical RH accuracy tolerances in the range of T = 0 °C … 80 °C are given in **Figure 6**, **Figure 7**, **Figure 8**, and **Figure 9**.**Figure 10**

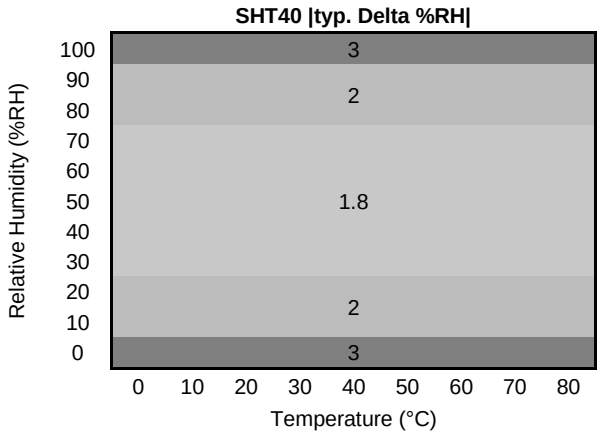


Figure 6. Typical RH accuracy tolerance over humidity and temperature for SHT40.

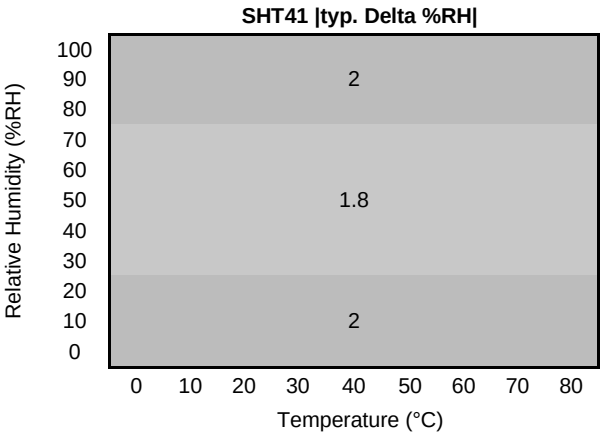


Figure 7. Typical RH accuracy tolerance over humidity and temperature for SHT41.

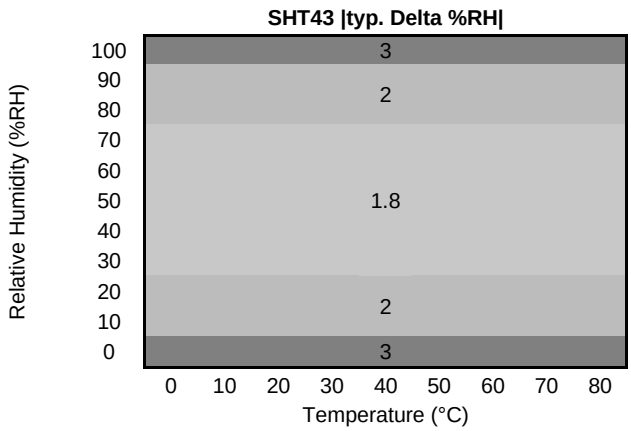


Figure 8. Typical RH accuracy tolerance over humidity and temperature for SHT43.

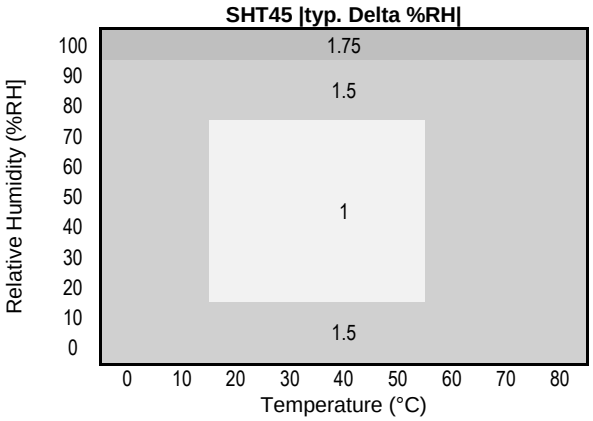


Figure 9. Typical RH accuracy tolerance over humidity and temperature for SHT45.

2.2 Temperature

Parameter	Conditions	Value	Units
SHT40 *T* Accuracy[1]	typ.	±0.2	°C
	max.	see **Figure 10**	-
SHT41 *T* Accuracy[1]	typ.	±0.2	°C
	max.	see **Figure 11**	-
SHT43 *T* Accuracy[1]	max.	see **Figure 12**	-
SHT45 *T* Accuracy[1]	typ.	±0.1	°C
	max.	see **Figure 13**	-
Repeatability[2]	high	0.04	°C
	medium	0.07	°C
	low	0.1	°C
Resolution[4]	-	0.01	°C
Specified range[5]	-	–40 to +125	°C
Response time[9]	$\tau_{63\%}$	2	s
Long-term drift[10]	typ.	<0.03	°C/y
Long-term drift for SHT43	typ.	<0.01	°C/y

Table 2. General temperature sensor specifications.

[9] Temperature response time depends on heat conductivity of sensor substrate and design-in of sensor in application.

[10] Max. value is <0.04 °C/y. And please note that it is different for the SHT43.

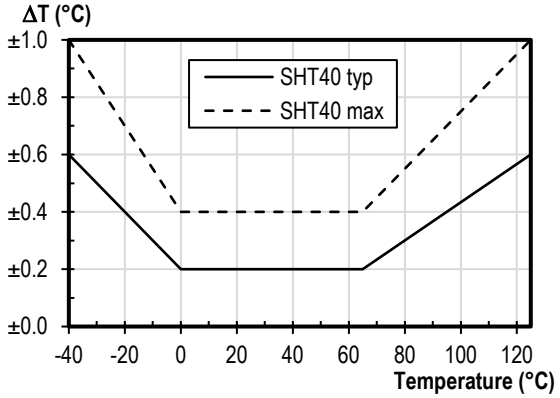


Figure 10. SHT40 typical and maximal temperature accuracy.

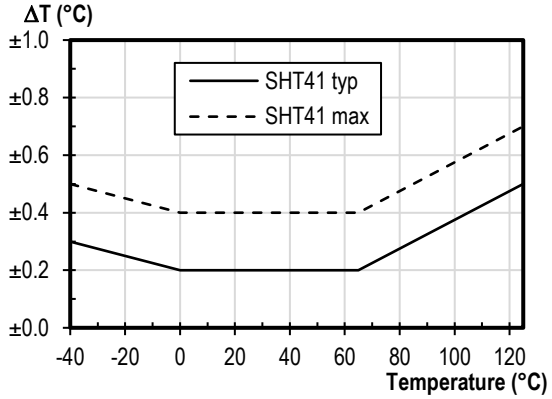


Figure 11. SHT41 typical and maximal temperature accuracy.

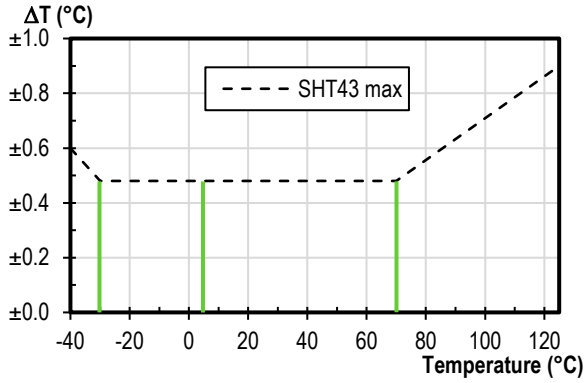


Figure 12. SHT43 maximal temperature accuracy. The green lines represent the calibration points for ISO-17025 certification.

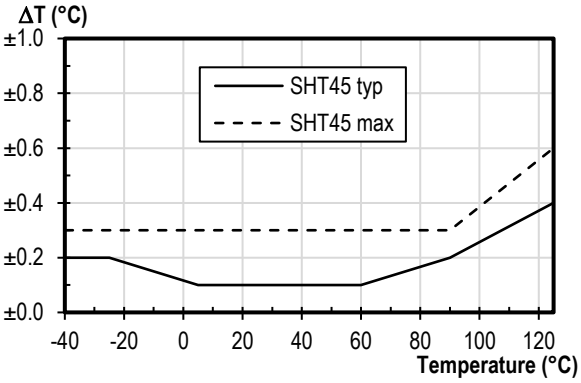


Figure 13. SHT45 typical and maximal temperature accuracy.

2.3 Recommended Operating Conditions

The sensor shows best performance when operated within the recommended normal temperature and humidity range of 5 °C … 60 °C and 20 %RH … 80 %RH, respectively. Long term exposure to conditions outside recommended normal range, especially at high relative humidity, may temporarily offset the RH signal (*e.g.* +3 %RH after 60 h at >80 %RH). After returning into the recommended normal temperature and humidity range the sensor will recover to within specifications by itself. Prolonged exposure to extreme conditions may accelerate ageing.

To ensure stable operation of the humidity sensor, the conditions described in the document “Handling Instructions for humidity Sensors” [2] regarding exposure to volatile organic compounds must be met. Please note as well that this does apply not only to transportation and manufacturing, but also to operation of the SHT4x.

2.4 ISO17025 certification with 3-point calibration data

All SHT43 can be uniquely identified by their serial number (read out command see paragraph 4.7). For each sensor an individual 3-point calibration is performed, accredited to ISO/IEC 17025:2017. The accreditation is performed and granted by the Swiss Accreditation Service (SAS), a public institution of the Swiss Government. The accreditation is documented on the SAS website under the name SCS 0158 and can be downloaded from this link[11]. The three calibration temperatures are T = −30 °C, T = 5 °C, and T = 70 °C. Measurement uncertainties and decision rules according to the SAS are given in **Table 3.**

Metrological traceability of the calibration is in accordance to ch. 6.5 of ISO/IEC 17025:2017, encompassing but not limited to NIST traceability or traceability to other national metrology institutes, according to the CIPM Mutual Recognition Arrangement (CIPM MRA).

Reel-wise calibration certificates and data for each SHT43 can be downloaded from libellus.sensirion.com. This allows for efficient processing by automated systems.

Temperature	Expanded measurement uncertainty (k=2)*	Decision rule
−30 °C	0.40 °C	Shared risk (JCGM 106:2010, 8.2)
5 °C	0.20 °C	Shared risk (JCGM 106:2010, 8.2)
70 °C	0.20 °C	Shared risk (JCGM 106:2010, 8.2)

Table 3. Measurement uncertainty and decision rule for the accredited calibration according to Swiss Accreditation Service (SAS). *Measurement uncertainties represent a confidence level of 95% using a coverage factor of k = 2.

[11] https://www.sas.admin.ch/sas/en/home/akkreditiertestellen/akkrstellensuchesas.exturl.html/aHR0cHM6Ly9zYXNkYi5jbGllbnRzLmxpaXAuY2gvc2VhcmNoLm/h0bWw=.html?csrfmiddlewaretoken=2le2f1aOTUge9YQ3nuaL0ILEYS980ZMdH60doLdY8Nh1sHC3wpw3YvLLVBezmDRn&lang=en&search_term=0158&accreditation_type=2&submit=Start+search

3 Electrical Specifications

Valid for all electrical specifications: Typical values correspond to V_{DD} = 3.3 V and T = 25 °C. Min. and max. values are valid in the full temperature range −40 °C … 125 °C and at declared V_{DD} levels and are based on characterization.

3.1 Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit	Comments
Supply voltage	V_{DD}		1.08	3.3	3.6	V	-
Power-up/down level	V_{POR}	Static power supply	0.6	-	1.08	V	-
Slew rate of the supply voltage	$V_{DD, slew}$		-	-	20	V/ms	Voltage changes on the supply between $V_{DD, min}$ and $V_{DD, max.}$ Faster slew rates may lead to a reset
Supply current (heater **off**)	I_{DD}	Idle state	- -	0.08 -	1.0 3.4	µA	At 25 °C At 125 °C
		Power up	-	50	-	µA	-
		Measurement	-	320	500	µA	Current while sensor is measuring
		Avg., high repeatability Avg., med. repeatability Avg., low repeatability	- - -	2.2 1.2 0.4	- - -	µA	Avg. current consumption (continuous operation with 1 meas. per second)
Supply current (heater **on**)	I_{DD}	Nomin. heater “200 mW”	-	60	100	mA	see section 4.9
		Nomin. heater “110 mW”	-	33	55	mA	
		Nomin. heater “20 mW”	-	6	10	mA	
Power consumpt. at VDD=1.2 V (no heater)	-	Avg., high repeatability Avg., med. repeatability Avg., low repeatability	- - -	2.6 1.4 0.5	- - -	µW	Avg. power consumption (continuous operation with 1 meas. per second)
Low level input voltage	V_{IL}	-	0	-	0.3* V_{DD}	V	-
High level input voltage	V_{IH}	-	0.7* V_{DD}	-	V_{DD}	V	-
Pull up resistors	R_p	V_{DD} < 1.62 V	820	-	-	Ω	-
		V_{DD} ≥ 1.62 V	390	-	-	Ω	-
Low level output voltage	V_{OL}	V_{DD} < 1.62 V, R_{pullup} > 820 Ω	-	-	0.2* V_{DD}	V	-
		V_{DD} = 1.62 V … 2.0 V, R_{pullup} > 390 Ω	-	-	0.2* V_{DD}	V	-
		V_{DD} > 2.0 V, R_{pullup} > 390 Ω	-	-	0.4	V	-
Capacitive bus load	C_b	R_P ≤ 820 Ω: fast mode	-	-	400	pF	Capacitive bus load can be determined from $C_b < t_{rise}/(0.8473*R_p)$. Rise times are t_{rise} = 300 ns for fast mode and t_{rise} = 120 ns for fast mode plus
		R_p = 390 Ω, VDD > 1.62 V: fast mode plus	-	-	340	pF	

Table 4. Electrical specifications.

3.2 Timings

Max. values are measured at −40 °C and 1.08 V supply voltage (based on characterization).

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Units	Comments
Power-up time	t_{PU}	After hard reset, $V_{DD} \geq V_{POR}$	-	0.3	1	ms	Time between V_{DD} reaching V_{POR} and sensor entering idle state
Soft reset time	t_{SR}	After soft reset	-	-	1	ms	Time between ACK of soft reset command and sensor entering idle state. Also valid for I2C general call reset.
Measurement duration	$t_{MEAS,l}$	Low repeatability	-	1.3	1.6	ms	Including t_{PU}: The three repeatability modes differ with respect to measurement duration, noise level and energy consumption
	$t_{MEAS,m}$	Med. repeatability	-	3.7	4.5	ms	
	$t_{MEAS,h}$	High repeatability	-	6.9	8.3	ms	
Heater-on duration	t_{Heater}	Long pulse	0.9	1	1.1	s	After that time the heater is automatically switched off
		Short pulse	0.09	0.1	0.11	s	After that time the heater is automatically switched off

Table 5. System timing specifications.

3.3 Absolute Maximum Ratings

Stress levels beyond those listed in **Table 6** may cause permanent damage or affect the reliability of the device. These are stress ratings only and functional operation of the device at these conditions is not guaranteed. Ratings are only tested each at a time.

Parameter	Rating
Max. voltage on any pin	V_{SS} −0.3 V … V_{DD} +0.3 V
Operating temperature range	−40 °C … 125 °C
Storage temperature range[12]	−40 °C …150 °C
ESD HBM	2 kV
ESD CDM	500 V
Latch up, JESD78 Class II, 125 °C	±100 mA

Table 6. Absolute maximum ratings.

[12] The recommended storage temperature range is 10-50 °C. Please consult the document “SHTxx Handling Instructions” [2] for more information.

4 Sensor Operation

4.1 I2C communication

I2C communication is based on NXP's I2C-bus specification and user manual UM10204 [3]. Supported I2C modes are standard, fast mode, and fast mode plus. Data is transferred in multiples of 16-bit words and 8-bit checksum (cyclic redundancy check = CRC). All transfers must begin with a start condition (S) and terminate with a stop condition (P). To finish a read transfer, send not acknowledge (NACK) and stop condition (P). Addressing a specific slave device is done by sending its 7-bit I2C address followed by an eighth bit, denoting the communication direction: "zero" indicates transmission to the slave, i.e. "write", a "one" indicates a "read" request. Schematics of the I2C transfer types are sketched in **Figure 14**. The sensor does not support clock-stretching. In case the sensor receives a read header and is still busy with *e.g.* measurement or heating, it will return a NACK. Measurement data can only be received once and will be deleted from the sensor's register after the first acknowledged I2C read header.

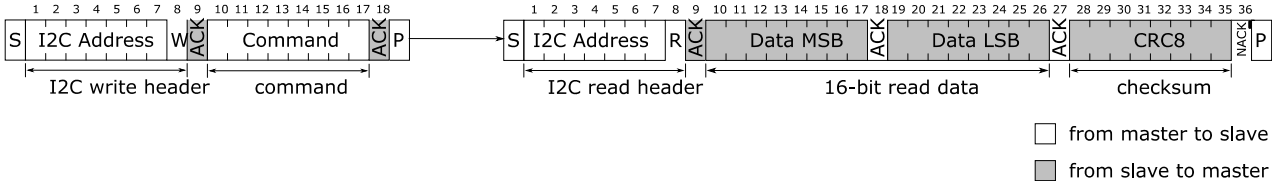


Figure 14. I2C transfer types: First a write header is sent to the I2C slave, followed by a command, for example "measure RH&T with highest precision". After the measurement is finished the read request directed to this I2C slave will be acknowledged and transmission of data will be started by the slave.

4.2 I2C Communication Timing

All details on the timing are following the interface specification of NXP's user manual UM10204 [3]. Please follow mandatory capacitor and resistor requirements given in **Table 4**.

4.3 Data type & length

I2C bus operates with 8-bit data packages. Information from the sensor to the master has a checksum after every second 8-bit data package.

Humidity and temperature data will always be transmitted in the following way: The first value is the temperature signal (2 * 8-bit data + 8-bit CRC), the second is the humidity signal (2 * 8-bit data + 8-bit CRC).

4.4 Checksum Calculation

For read transfers each 16-bit data is followed by a checksum with the following properties

Property	Value
Name	CRC-8
Message Length	16-bit
Polynomial	0x31 ($x^8 + x^5 + x^4 + 1$)
Initialization	0xFF
Reflect Input/Output	false/false
Final XOR	0x00
Examples	CRC(0xBEEF) = 0x92

Table 7. Data checksum properties.

The master may abort a read transfer after the 16-bit data if it does not require a checksum.

4.5 Command Overview

Command (hex)	Response length incl. CRC (bytes)	Description [return values]
0xFD	6	measure T & RH with high precision (high repeatability) [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0xF6	6	measure T & RH with medium precision (medium repeatability) [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0xE0	6	measure T & RH with lowest precision (low repeatability) [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x89	6	read serial number [2 * 8-bit data; 8-bit CRC; 2 * 8-bit data; 8-bit CRC]
0x94	-	soft reset [ACK]
0x39	6	activate heater with 200mW for 1s, including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x32	6	activate heater with 200mW for 0.1s including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x2F	6	activate heater with 110mW for 1s including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x24	6	activate heater with 110mW for 0.1s including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x1E	6	activate heater with 20mW for 1s including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]
0x15	6	activate heater with 20mW for 0.1s including a high precision measurement just before deactivation [2 * 8-bit T-data; 8-bit CRC; 2 * 8-bit RH-data; 8-bit CRC]

Table 8. Overview of I2C commands. If the sensor is not ready to process a command, *e.g.* because it is still measuring, it will response with NACK to the I2C read header. Given heater power values are typical and valid for VDD=3.3 V.

4.6 Conversion of Signal Output

The digital sensor signals correspond to following humidity and temperature values:

$$RH = \left(-6 + 125 \cdot \frac{S_{RH}}{2^{16}-1}\right) \%\text{RH} \quad (1)$$

$$T = \left(-45 + 175 \cdot \frac{S_T}{2^{16}-1}\right) °\text{C} \quad (2)$$

$$T = \left(-49 + 315 \cdot \frac{S_T}{2^{16}-1}\right) °\text{F} \quad (3)$$

N.B.: The RH conversion formula (1) allows values to be reported which are outside of the range of 0 %RH … 100 %RH. Relative humidity values which are smaller than 0 %RH and larger than 100 %RH are non-physical, however these “uncropped” values might be found beneficial in some cases (*e.g.* when the distribution of the sensors at the measurement boundaries are of interest). For all [3]s who do not want to engage in evaluation of these non-physical values, cropping of the RH signal to the range of 0 %RH … 100 %RH is advised.

4.7 Serial number

Each sensor has a unique serial number, that is assigned by Sensirion during production. It is stored in the one-time-programmable memory and cannot be manipulated after production. The serial number is accessible via I2C command 0x89 and is transmitted as two 16-bit words, each followed by an 8-bit CRC.

4.8 Reset & Abort

A reset of the sensor can be achieved in three ways:

1. Soft reset: send the reset command described in **Table 8**.
2. I2C general call reset: all devices on I2C bus are reset by sending the command 0x06 to the I2C address 0x00.
3. Power down (incl. pulling SCL and SDA low)

Any command that triggers an action at the sensor can be aborted via I2C general call reset or soft reset.

4.9 Heater Operation

The sensor incorporates an integrated on-package heater which can be switched on by the set of commands given in **Table 8**. Three heating powers and two heating durations are selectable. After reception of a heater-on command, the sensor executes the following procedure:

1. The heater is enabled, and the timer starts its count-down.
2. On timer expiration a temperature and humidity measurement with the highest repeatability is started, the heater remains enabled.
3. After the measurement is finished the heater is turned off.
4. Temperature and humidity values are now available for readout.

The maximum on-time of the heater commands is one second in order to prevent overheating of the sensor by unintended usage of the heater. Thus, there is no dedicated command to turn off the heater. For extended heating periods it is required to send periodic heater-on commands, keeping in mind that the heater is designed for a maximal duty cycle of less than 10%. To obtain a fast increase in temperature the idle time between consecutive heating pulses shall be kept minimal.

Possible Heater Use Cases

There will be dedicated Sensirion application notes elaborating on various use cases of the heater. In general, the applications of the on-package heater range around:

1. Removal of condensed / spray water on the sensor surface. Although condensed water is not a reliability / quality problem to the sensor, it will however make the sensor non-responsive to RH changes in the air as long as there is liquid water on the surface.
2. Creep-free operation in high humid environments. Periodic heating pulses allow for creep-free high-humidity measurements for extended times.

Important notes for operating the heater:

1. The heater is designed for a maximum duty cycle of 10%, meaning the total heater-on-time should not be longer than 10% of the sensor's lifetime.
2. During operation of the heater, sensor specifications are not valid.
3. The temperature sensor can additionally be affected by the thermally induced mechanical stress, offsetting the temperature reading from the actual temperature.
4. The sensor's temperature (base temperature + temperature increase from heater) must not exceed T_{max} = 125 °C in order to have proper electrical functionality of the chip.
5. The heater draws a large amount of current once enabled (up to ~75 mA in the highest power setting). Although a dedicated circuitry draws this current smoothly, the power supply must be strong enough to avoid large voltage drops that could provoke a sensor reset.
6. If higher heating temperatures are desired, consecutive heating commands have to be sent to the sensor. The heater shall only be operated in ambient temperatures below 65 °C else it could drive the sensor outside of its maximal operating temperature.

5 Physical Specification

5.1 Package Description

SHT4x is provided in an open-cavity dual flat no lead (DFN) package. The humidity sensor opening is centered on the top side of the package. The sensor chip is made of silicon, hosted on a copper lead frame and overmolded by an epoxy-based mold compound. Exposed bottom side of the leadframe with the metallic contacts is Ni/Pd/Au coated, side walls are bare copper.
Moisture sensitivity level (MSL) of one according to IPC/JEDEC J-STD-020 is achieved. It is recommended to process the sensors within one year after date of delivery.

5.2 Package Outline

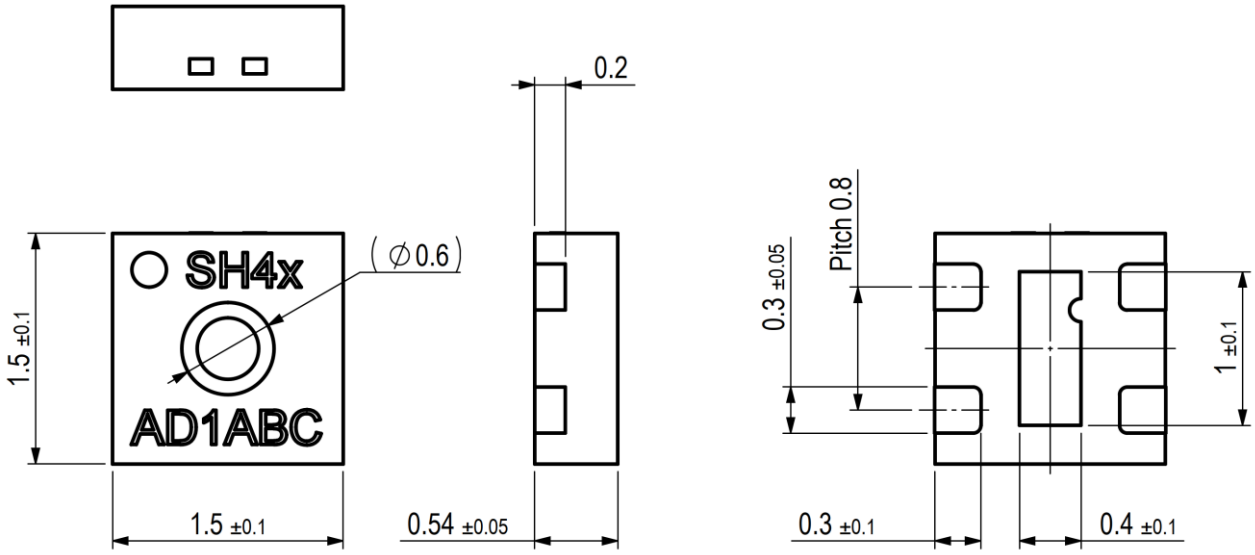


Figure 15. Dimensional drawing of SHT4x including package tolerances (units mm).

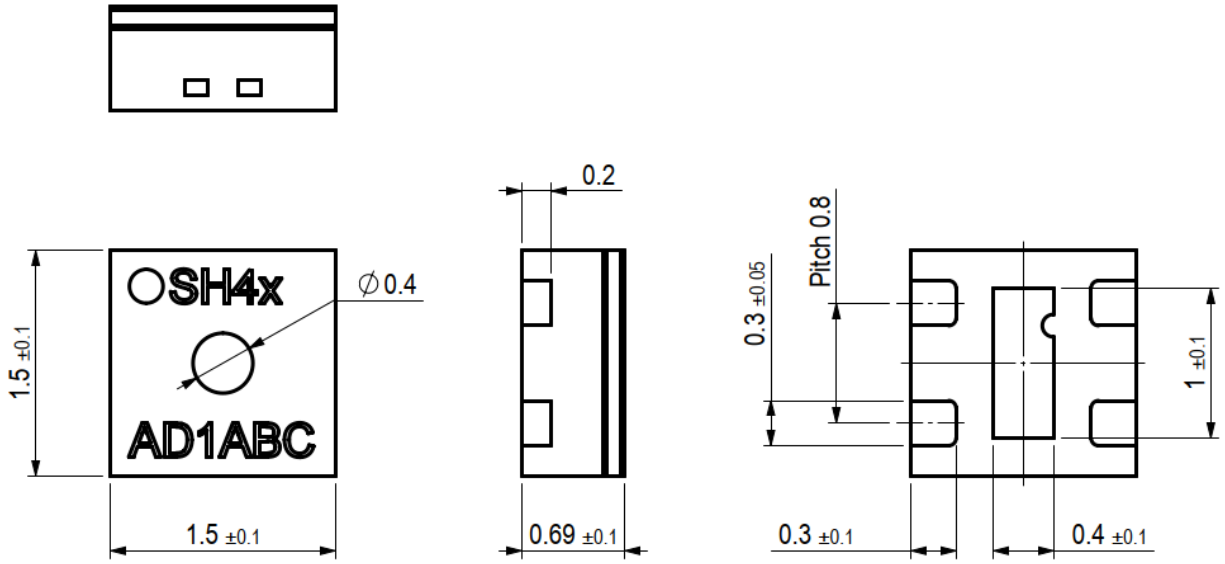


Figure 16. Dimensional drawing of SHT4xI with filter membrane including package tolerances (units mm)

5.3 Land Pattern

The land pattern is recommended to be designed according to the used PCB and soldering process together with the physical outer dimensions of the sensor. For reference, the land pattern used with Sensirion's PCBs and soldering processes is given in **Figure 17**. Soldering of the central die pad is optional. Sensirion recommends to not solder the central die pad because the sensor can reach higher temperatures upon heater activation

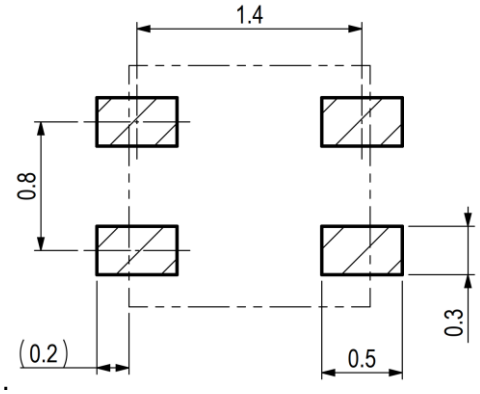


Figure 17. Recommended land pattern (in mm). Details can vary and depend on used PCBs and solder processes. There shall be no copper under the sensor other than at the pin pads.

5.4 Pin Assignment & Laser Marking

Pin	Name	Comments
1	SDA	Serial data, bidirectional
2	SCL	Serial clock, unidirectional input
3	VDD	Supply voltage
4	VSS	Ground



Figure 18. Pin assignment (transparent top view). Dashed lines are only visible if sensor is viewed from below. The die pad is not directly connected to any pin.

The laser marking consists of two lines, indicated in **Figure 18**. In the first line a filled circle serves as pin-1 indicator and is followed by "SH4". The last character will indicate the accuracy class of this product (here "x" serves as place holder). In the second line, the first three characters specify the product characteristics according to positions 7, 8 and 9 of **Table 10**. The second three characters serve as internal batch tracking code.[13]

[13] Please note, there will be no change in the laser marking for the protective option (filter membrane and protective cover).

5.5 Thermal Information

Symbol	Description	Heater off, die pad soldered (K/W)	Heater on, die pad soldered (K/W)	Heater off, **die pad not soldered** (K/W)	Heater on, **die pad not soldered** (K/W)
$R_{\theta JA}$	Junction-to-ambient thermal resistance	246	308	**297**	**357**
$R_{\theta JC}$	Junction-to-case thermal resistance	189	255	**191**	**257**
$R_{\theta JB}$	Junction-to-board thermal resistance	159	225	**193**	**258**
Ψ_{JB}	Junction-to-board characterization param.	159	223	**191**	**254**
Ψ_{JT}	Junction-to-top characterization param.	38	105	**44**	**112**

Table 9. Typical values for thermal metrics. In the “heater on” columns a heater power of 200 mW was assumed. Soldering of the die pad is not recommended, therefore the two right hand side columns are bold. Values are based on simulation.

6 Protection Options

6.1 Membrane Option

The filter membrane option for SHT4x family members inherently provides an additional barrier for all pollutants to enter the sensor opening, thus lowering negative influences on the sensing element. Mostly designed to keep particles and dust from accumulating and reducing the response time, the membrane also enables more efficient and easy cleaning, as it helps to reduce liquid intrusion into the sensor opening. Even though not selectively filtering, in general, physical barriers allow to reduce the amount of unwanted chemical contamination and help to remove potentially harmful components by facilitating wiping (flat sensor surface).

The integrated SHT4x PTFE membrane provides additional protection, from particles, and enables sensor operation in harsh conditions. The membrane has a thickness of 100 µm offering a filtration efficiency of >99.99% for particles of 200 nm size and larger. Owing to the high permeability and the small volume between sensing element and membrane, the specified response time of the RH sensor is unaltered.



Figure 19. SHT4x with integrated PTFE membrane, highlighting the beneficial flat geometry of the SHT4xI.

To ensure full functionality of the sensor and avoid damaging its integrated filter membrane, when mounting the sensor, follow the reflow soldering process as described in the Handling Instructions [2]. Furthermore the therein described care regarding board wash and cleaning still apply.

6.2 Protective Cover

The SHT4x will be available with a second protective option, a removable protective cover to protect the sensing element during sensor installation. The sensor will be delivered with the protective cover attached such that the sensor opening is completely covered and sealed. This enables cost-effective brush-over and spray-over application procedures of conformal coating material. Such coating is often required in highly corrosive environments to protect solder joints. In this process the protective cover prevents the sensor opening to be sealed by any coating and to it, the foil can be pulled off with tweezers at the designated non-sticking flap.

The protective cover is made of polyimide making it highly resistant to chemicals and elevated temperatures[14]. To ensure full functionality of the cover, when mounting the sensor, follow the reflow soldering process as described in the Handling Instruction [2].

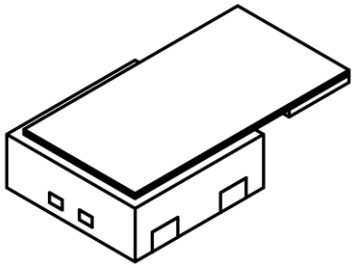


Figure 20. Sketch of the SHT4x with attached polyimide foil.[15]

7 Quality and Material Contents

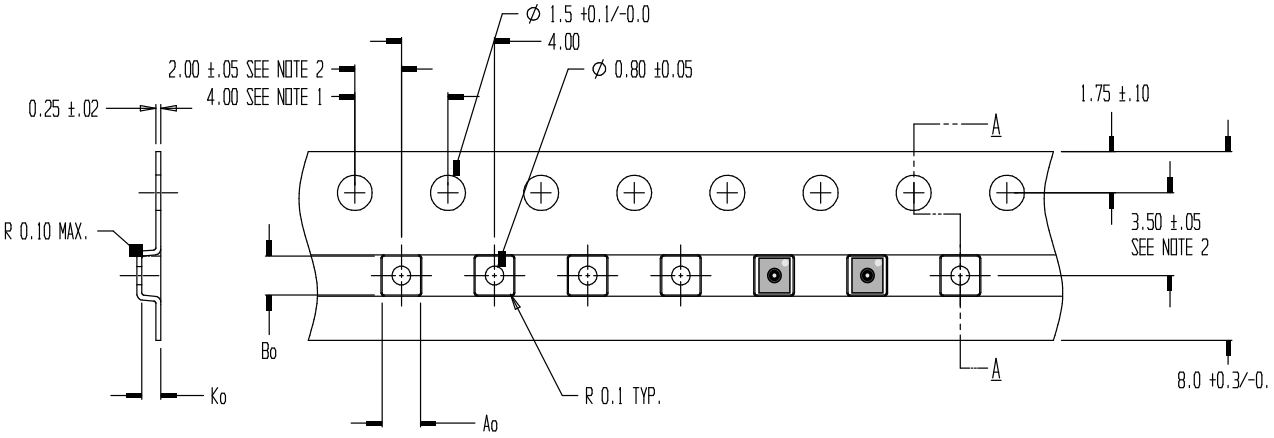
Qualification of SHT4x is performed based on the JEDEC JESD47 qualification test method, qualification report available on request. The device is fully RoHS and WEEE compliant, *e.g.* free of Pb, Cd, and Hg.
For general remarks of best practice in processing humidity sensor please refer to the handling instructions [2].

[14] Up to 260 °C

[15] This a preliminary sketch only and might be subject to change.

8 Tape and Reel Packaging

All specifications for the tape and reel packaging can be found on **Figure 21**. Reel diameters are 13 inch and 8 inch for the 10k and the 2.5k packaging sizes, respectively.

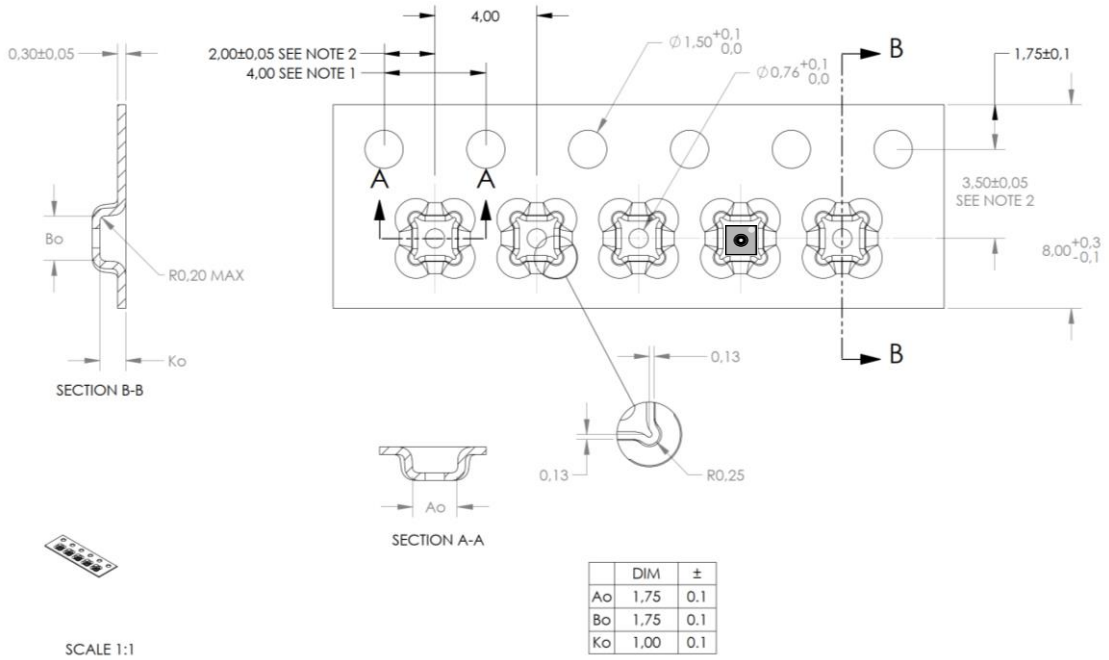


NOTES:
1. 10 SPROCKET HOLE PITCH CUMULATIVE TOLERANCE ±0.2
2. POCKET POSITION RELATIVE TO SPROCKET HOLE MEASURED AS TRUE POSITION OF POCKET, NOT POCKET HOLE
3. Ao AND Bo ARE CALCULATED ON A PLANE AT A DISTANCE "R" ABOVE THE BOTTOM OF THE POCKET.

Ao = 1.65 ±0.05
Bo = 1.65 ±0.05
Ko = 0.81 ±0.05

TOLERANCES - UNLESS NOTED 1PL ±.2 2PL ±.10

Figure 21. Tape and reel specifications including sensor orientation in pocket (see indication of two sensors on the right side of the tape).



	DIM	±
Ao	1,75	0.1
Bo	1,75	0.1
Ko	1,00	0.1

NOTES:
1. 10 SPROCKET HOLE PITCH CUMULATIVE TOLERANCE ±0.2
2. POCKET POSITION RELATIVE TO SPROCKET HOLE MEASURED AS TRUE POSITION OF POCKET, NOT POCKET HOLE.
3. Ao AND Bo ARE MEASURED ON A PLANE AT A DISTANCE "R" ABOVE THE BOTTOM OF THE POCKET.

Figure 22. Tape and reel specification including sensor orientation in pocket of sensor with membrane option.

9 Product Name Nomenclature

Position	Value(s)	Explanation
1	S	Sensirion
2	H	Humidity Signal
3	T	Temperature Signal
4	4	Fourth product generation
5	0 1 5 3	Base accuracy Intermediate accuracy Best accuracy ISO17025 certified
6	-	delimiter
7	A B C	I2C interface with 0x44 address I2C interface with 0x45 address I2C interface with 0x45 address
8	D	DFN package
9	1 C	Reserved 3-point calibrated and certified
10	B F P	Blank package Package with integrated, patented PTFE membrane Package with removable protective cover for conformal coating (coming soon)
11	-	delimiter
12	R	Tape on reel packaging
13	2 3	Packaging article contains 2’500 pieces Packaging article contains 10’000 pieces

Table 10. SHT4x product name nomenclature.

10 Ordering Information

Material Description	Material Number	Details	Quantity (pcs)
SHT40-AD1B-R2	3.000.465	base RH&T acc., 0x44 I2C addr.	2’500
SHT40-AD1B-R3	3.000.353	base RH&T acc., 0x44 I2C addr.	10’000
SHT40-AD1F-R2	3.000.820	base RH&T acc., 0x44 I2C addr., including patented PTFE membrane	2’500
SHT40-AD1P-R2	tbd	base RH&T acc., 0x44 I2C addr., including removable protective cover **available Q2/23**	2’500
SHT40-BD1B-R2	3.000.492	base RH&T acc., 0x45 I2C addr.	2’500
SHT40-BD1F-R2	3.000.887	base RH&T acc., 0x45 I2C addr.	2’500
SHT40-BD1B-R3	3.000.610	base RH&T acc., 0x45 I2C addr.	10’000
SHT40-CD1B-R3	3.000.691	base RH&T acc., 0x46 I2C addr.	10’000
SHT41-AD1B-R2	3.000.466	intermed. RH&T acc., 0x44 I2C addr.	2’500
SHT41-AD1B-R3	3.000.611	intermed. RH&T acc., 0x44 I2C addr.	10’000
SHT41-AD1F-R2	3.000.885	intermed. RH&T acc., 0x44 I2C addr including patented PTFE membrane	2’500
SHT43-ADCB-R2	3.000.682	3-point calibrated, ISO17025 certified, 0x44 I2C addr.	2’500
SHT43-ADCB-R3	3.000.823	3-point calibrated, ISO17025 certified, 0x44 I2C addr.	10’000
SHT43-BDCB-R3	3.000.904	3-point calibrated, ISO17025 certified, 0x45 I2C addr.	10’000
SHT45-AD1B-R2	3.000.645	±1.0 %RH, ±0.1 °C acc., 0x44 I2C addr.	2’500
SHT45-AD1F-R2	3.000.886	±1.0 %RH, ±0.1 °C acc., 0x44 I2C addr. Including patented PTFE membrane	2’500
SHT45-AD1B-R3	3.000.750	±1.0 %RH, ±0.1 °C acc., 0x44 I2C addr.	10’000

Table 11. SHT4x ordering options.

11 Bibliography

[1] K. Ehrhorn, "A Humidity Sensor and a Methode for Manufacturing the Same.". UK, DE, FR, NL, DM, BE, US, CN Patent EP1810013, US7741950, CN101040181, 2005.

[2] Sensirion, "Handling Instructions for Humidity Sensors," 2020.

[3] NXP Semiconductors, "User manual UM10204," vol. Rev. 6, 2014.

12 Revision History

Date	Version	Page(s)	Changes
October 2020	1	all	Initial release
July 2021	2	multiple 3 4 4 11 10 13 16 12 13 13 20	Typo correction Included checksum in **Figure 1** Included description of NIST traceability in section 2 Included repeatability clarification in **Table 1** Clarified I2C communication in section 4.1 Removed waiting time specification in **Table 5** Specified serial number in 4.7 Updated qualification status in section 6 Deleted binary com. & included return values in **Table 8** Updated note on duty cycle of heater in section 4.9 Added note on large current drawn by heater in section 4.9 Updated ordering information in **Table 11**
March 2022	3	multiple multiple 4 4 4 10 11 19 20	Included SHT45 RH- and T-accuracy specifications Extended max. heater duty cycle to 10% Reduced RH response time to 4s in **Table 1** Reduced long-term drift to <0.2 %RH/y in **Table 1** Reduced hysteresis to 0.8 %RH at 25 °C in **Table 1** Updated max. measurement times in **Table 5** Included I2C communication timing in section 4.2 Introduced new product version in **Table 10** Updated ordering information in **Table 11**
November 2022	4	all 1 4 5 6 7 7 8 14 15 16 15 18 19 20 All	Updated Datasheet with new SHT43 New Features and General Description and Device Overview Edited **Table 1** with SHT43 data Inserted **Figure 4.** SHT43 typical and maximal relative humidity accuracy at 25 °C. Edited **Table 2** with Sht43 data **Table 2.** General temperature sensor specifications. Inserted **Figure 8.** Typical RH accuracy tolerance over humidity and temperature for SHT43 Inserted **Figure 12** for SHT 43 Inserted section 2.4 ISO17025 certification with 3-point calibration data Inserted **Figure 16** for filter membrane Added footnote 13 for Laser marking of protective options Inserted section 6 about protective options Inserted **Figure 20** showing protective cover Inserted **Figure 22.** Tape and reel specification including sensor orientation in pocket of sensor with membrane option. Updated Product Name Nomenclature with SHT43 (3 & C) Updated Ordering Information with SHT43-ABDC-R3 Reformatting
January 2023	5	20	Added SHT43-BDCB-R3 to **Table 11**

Important Notices

Warning, Personal Injury

Do not use this product as safety or emergency stop devices or in any other application where failure of the product could result in personal injury. Do not use this product for applications other than its intended and authorized use. Before installing, handling, using or servicing this product, please consult the data sheet and application notes. Failure to comply with these instructions could result in death or serious injury.

If the Buyer shall purchase or use SENSIRION products for any unintended or unauthorized application, Buyer shall defend, indemnify and hold harmless SENSIRION and its officers, employees, subsidiaries, affiliates and distributors against all claims, costs, damages and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if SENSIRION shall be allegedly negligent with respect to the design or the manufacture of the product.

ESD Precautions

The inherent design of this component causes it to be sensitive to electrostatic discharge (ESD). To prevent ESD-induced damage and/or degradation, take customary and statutory ESD precautions when handling this product. See application note "ESD, Latchup and EMC" for more information.

Warranty

SENSIRION warrants solely to the original purchaser of this product for a period of 12 months (one year) from the date of delivery that this product shall be of the quality, material and workmanship defined in SENSIRION's published specifications of the product. Within such period, if proven to be defective, SENSIRION shall repair and/or replace this product, in SENSIRION's discretion, free of charge to the Buyer, provided that:

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- such defects shall be found, to SENSIRION's reasonable satisfaction, to have arisen from SENSIRION's faulty design, material, or workmanship;
- the defective product shall be returned to SENSIRION's factory at the Buyer's expense; and
- the warranty period for any repaired or replaced product shall be limited to the unexpired portion of the original period.

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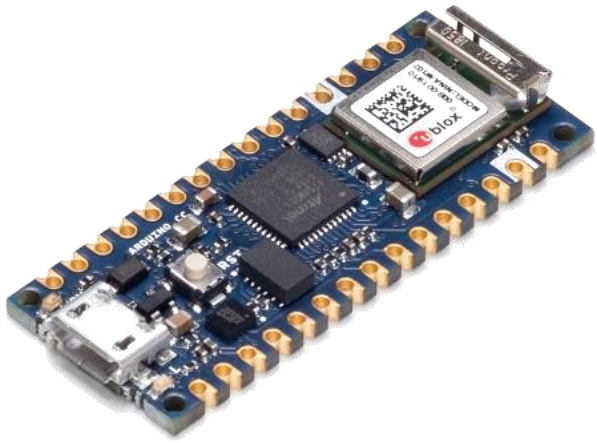
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ABX00027: PLACA DE DESARROLLO ARDUINO NANO 33 IOT

Distribuidor ARDUINO autorizado



Descripción

El ABX00027 es una placa de desarrollo llamada Arduino Nano 33 IoT, es pequeña, robusta y potente tiene conectividad WiFi y Bluetooth. El Arduino Nano 33 IoT es compatible con Arduino IoT Cloud y es compatible con el transporte seguro TLS completo: el cryptochip ATECC608A almacena las claves criptográficas en el hardware, ofreciendo un nivel muy alto de seguridad. La integración con Arduino IoT Cloud también ofrece una forma eficiente de configurar cuadros de mando con poca codificación.

Con el mismo tamaño del Arduino Nano, el Arduino Nano 33 IoT tiene un procesador Arm Cortex-M0 + SAMD21, un módulo WiFi, Bluetooth basado en ESP32, una Unidad de Medición Inercial (IMU) de 6 ejes y un chip criptográfico, que puede almacenar de forma segura certificados y claves pre compartidas.

Aplicaciones

El Arduino Nano 33 IoT se puede utilizar en un protoboard al colocarle pin header, o como SMT soldado directamente en una PCB. Se utiliza como herramienta educativa para aprender sobre el desarrollo de aplicaciones, agregando conectividad WiFi y Bluetooth con una fuerte seguridad a todas las aplicaciones.

El IMU a bordo se puede usar para monitorear de forma inalámbrica la vibración, la orientación y la velocidad de rotación de los objetos pequeños gracias a su factor de forma liviano y compacto. Funciona directamente con altos voltajes de las baterías de iones de litio de plomo o de varias celdas que proporcionan una fuente de alimentación de 3.3 V a los periféricos a una corriente de salida significativa.

Detalles del producto

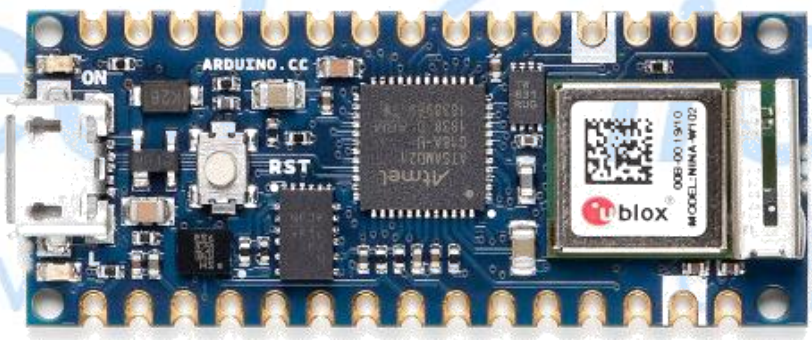
Esta placa está basada en el microcontrolador SAMD21G18A.

DETALLES	
Reloj	Hasta 48MHz
Flash	256KB
SRAM	32KB

Tenga en cuenta que Arduino Nano 33 IoT solo admite 3.3V en entradas/salidas y NO es tolerante a 5V, por lo tanto, asegúrese de que la placa no está conectada directamente señales de 5V, ya que de lo contrario se dañará. Además, a diferencia de las placas Arduino Nano que admiten el funcionamiento de 5 V, el pin de 5V NO suministra voltaje, sino que está conectado, a través de un puente, a la entrada de alimentación USB.

Para evitar riesgo de daños con proyectos existentes, donde debería poder sacar un Nano y reemplazarlo con el nuevo Nano 33 IoT, tenemos el pin de 5V en el encabezado, ubicado entre RST y A7 que no está conectado como configuración predeterminada de fábrica . Esto significa que si tiene un diseño que toma 5V de ese pin, no funcionará de inmediato.

El pin 5V está disponible solo cuando se cumplen dos condiciones: hacer un puente de soldadura en los dos pines marcados como V_{USB} y se alimenta el Nano 33 IoT a través del puerto USB. Si alimenta la placa desde el pin VIN, no obtendrá 5V regulados, por lo tanto, si hace el puente de soldadura, no saldrá nada de ese pin de 5V.



La comunicación en WiFi y Bluetooth es administrada por un módulo basado en NINA W102 ESP32. El módulo está conectado al microcontrolador SAMD21 con un bus SPI y un puerto serie a través de los siguientes pines:

PIN SAMD21	SAMD21 ACRÓNIMO	PIN NINA	NINA ACRÓNIMO	DESCRIPCIÓN
13	PA8	19	RESET_N	Reinicio
39	PA27	27	GPIO0	Solicita atención
41	PA28	7	GPIO33	Reconocer
23	PA14	28/21	GPIO5/ GPIO19	SPI CS / UART RTS
24	PA15	29/20	GPIO18/ GPIO22	SPI CLK / UART CTS
22	PA13	1	GPIO21	SPI MISO
21	PA12	36	GPIO12	SPI MOSI
31	PA22	23	GPIO3	Procesador TX -> Nina RX
32	PA23	22	GPIO1	NINA TX -> Procesador RX

Algunos de los pines NINA W102 están conectados a los encabezados / almohadillas de 15 + 15 pines y pueden ser controlados directamente por el módulo ESP32; en este caso, es necesario que los pines correspondientes de SAMD21 estén acertadamente tri-establecidos. A continuación se muestra una lista de tales señales:

PIN SAMD21	SAMD21 ACRÓNIMO	PIN NINA	NINA ACRÓNIMO	DESCRIPCIÓN DEL HEADER
48	PB03	8	RESET_N	A7
14	PA09	5	GPIO0	A6
8	PB09	31	GPIO33	AS/SCL
7	PB08	35	GPIO5/GPIO19	A4/SDA

La IMU es una LSM6DSL y se administra a través de I2C.

El chip criptográfico es un ATECC608A y tiene una biblioteca de soporte que es utilizada por la biblioteca WiFiNINA.

La placa tiene dos conectores de 15 pines (uno en cada lado), pin a pin compatible con la placa de desarrollo Arduino Nano.

PIN	FUNCION	TIPO	DESCRPCION
1	D13	Digital	GPIO
2	+3V3	Salida de VCC	Potencia generada internamente a dispositivos externos.
3	AREF	Analógica	Referencia analógica; puede ser utilizado como GPIO
4	AO/DACO	Analógica	ADC entrada / DAC salida; puede ser utilizado como GPIO
5	A1	Analógica	ADC entrada; puede ser utilizado como GPIO
6	A2	Analógica	ADC entrada; puede ser utilizado como GPIO
7	A3	Analógica	ADC entrada; puede ser utilizado como GPIO
8	A4/SDA	Analógica	ADC entrada; I 2 C SDA; Puede ser utilizado como GPIO(*)
9	A5/SCL	Analógica	ADC entrada; I 2 C SDA; Puede ser utilizado como GPIO(*)
10	A6	Analógica	ADC entrada; puede ser utilizado como GPIO
11	A7	Analógica	ADC entrada; puede ser utilizado como GPIO
12	V_{USB}	Entrada / Salida de energía	Normalmente NC; se puede conectar a la clavija USB V del conector USB cortando un puente
13	RST	Entrada digital	Entrada de restablecimiento bajo activo (duplicado del pin 18)
14	GND	Tierra	Tierra
15	VIN	Entrada de VCC	Entrada de energía
16	TX	Digital	USART TX; puede ser utilizado como GPIO
17	RX	Digital	USART RX; puede ser utilizado como GPIO
18	RST	Digital	Entrada de restablecimiento bajo activo (duplicado del pin 13)
19	GND	Tierra	Tierra
20	D2	Digital	GPIO
21	D3/PWM	Digital	GPIO; se puede utilizar como PWM
22	D4	Digital	GPIO
23	D5/PWM	Digital	GPIO; se puede utilizar como PWM
24	D6/PWM	Digital	GPIO; se puede utilizar como PWM
25	D7	Digital	GPIO
26	D8	Digital	GPIO
27	D9/PWM	Digital	GPIO; se puede utilizar como PWM
28	D10/PWM	Digital	GPIO; se puede utilizar como PWM
29	D11/MOSI	Digital	SPI MOSI; puede ser utilizado como GPIO
30	D12/MISO	Digital	SPI MOSI; puede ser utilizado como GPIO

(*) A diferencia de otras placas Arduino Nano, los pines A4 y A5 tienen un pull up interno y se utilizan por defecto como bus I^2C, por lo que no se recomienda su uso como entradas analógicas.

En la parte inferior de la placa, debajo del módulo de comunicación, las señales de depuración están organizadas como almohadillas de prueba 3x2. El pin 1 esta la parte inferior izquierda con el conector USB y las almohadillas de prueba a la derecha.

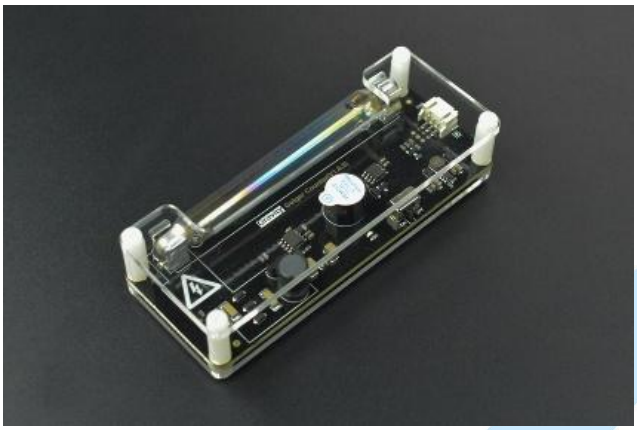
PIN	FUNCION	TIPO	DESCRPCION
1	+3V3	Salida de VCC	Salida de potencia generada internamente para ser utilizada como referencia de voltaje
2	SWD	Digital	Datos de depuración de un solo cable SAMD11
3	SWCLK	Entrada digital	Reloj de depuración de un solo cable SAMD11
4	UPDI	Digital	Interfaz de actualización ATMega4809
5	GND	Tierra	Tierra
6	RST	Entrada digital	Entrada de restablecimiento bajo activo



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TOLERANCIA: N/A	PLACA DE DESARROLLO ARDUINO NANO 33 IOT		
TOLERANCIA: N/A	Fecha: 18/06/2019	**No. Parte: ABX00027**	

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SEN0463
DETECTOR DE RADIACIÓN IONIZANTE (CONTADOR GEIGER)



DESCRIPCIÓN:

La radiación ionizante, un enemigo invisible e intangible, existe no sólo en los reactores de las centrales nucleares. De hecho, somos bombardeados por la radiación del entorno y del espacio exterior todo el tiempo, pero afortunadamente nuestro cuerpo es lo suficientemente fuerte como para resistir la radiación natural de fondo.

CARACTERISTICAS:

La ausencia de contacto activo no significa que no se encontrarán radiaciones ionizantes de alta energía. Los materiales de construcción de mármol natural, las gemas minerales de diferentes colores y el "polvo de iones negativos" de composición desconocida pueden contener diferentes cantidades de elementos radiactivos. Con el uso de un contador Geiger, estas fuentes radiactivas no tienen dónde esconderse.

Además, el contador Geiger es un buen generador de números aleatorios, y los eventos indeterminados de ionización de partículas de alta energía pueden proporcionar suficiente entropía aleatoria para obtener un número verdaderamente aleatorio, en lugar de una secuencia aleatoria fija basada en un algoritmo aleatorio.

ESPECIFICACIONES:

- Tensión de alimentación: 3.3 V ~ 5 V.
- Salida de señal: Salida digital, tira hacia abajo cuando se detecta un pulso.
- Voltaje de conducción: ≈400V.
- Rango máximo: 1200 μSv/h (valor teórico).
- Dimensiones: 107 mm × 42 mm.

TUBO GEIGER M4011:

- Voltaje de funcionamiento: 380V ~ 450V.
- Recuento de fondo: ≈25CPM.
- Relación CPM: 153,8 CPM/(μSv/h).
- Dimensiones: Φ10mm × 88mm.

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Realizó	LFSR
Revisó	ARSL
Fecha	17/11/2022



PINOUT:
1 - Fuente de alimentación negativa
2 + Fuente de alimentación positiva
3 Salida de señal D
4 Interruptor de señal acústica

NOTAS:

- Este producto no es un instrumento de medición profesional, solo es adecuado para investigación de principios y demostración de enseñanza. No apto para uso en mediciones de dosis de radiación que afecten directamente a la seguridad personal.
- ⚠ El tubo Geiger es frágil, manipúlelo con cuidado cuando lo utilice.
- ☢ El contador Geiger solo puede detectar radiación ionizante, como radiación nuclear, rayos X, rayos cósmicos, etc. No puede detectar radiación electromagnética, como radiación de hornos microondas, radiación de teléfonos móviles, radiación WiFi, radiación de cocinas de inducción.
- ☢ El tubo Geiger M4011 puede detectar rayos α, rayos β y rayos γ. Entre ellos, los rayos β duros y los rayos γ se pueden medir cuantitativamente.
- ☀ La exposición a una luz intensa puede provocar lecturas altas en el contador Geiger; evite la luz solar directa sobre el tubo Geiger.
- ⚡ El tubo Geiger funciona con un voltaje de hasta 400 V; después de encenderlo, no toque el circuito de alto voltaje cerca del polo positivo del tubo Geiger.

Tutorial y ejemplo:

https://wiki.dfrobot.com/SKU_SEN0463_Gravity_Geiger_Counter_Module

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